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Soham



## Content Management System

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# Content Management Systems

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## Creating a blog landing page

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# Content Management Systems

## Creating a blog landing page

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The Views module in Drupal is a visual query builder that enables you to build dynamic content displays without writing any code.

We will walk through how to create a page to list blogs and a companion block to display five of the most recent blogs. Then, we will move into creating exposed filters to allow end users to control the view results .

# Content Management Systems

## Creating a blog landing page

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### **Creating a blog landing page**

The Views module does one thing, and it does it well – listing content. The power behind the Views module is the amount of configurable power it gives to end users to display content in various forms.

This step will cover the process of how to create a content listing and link it to the main menu. We will use the Article content type to create a blog landing page.

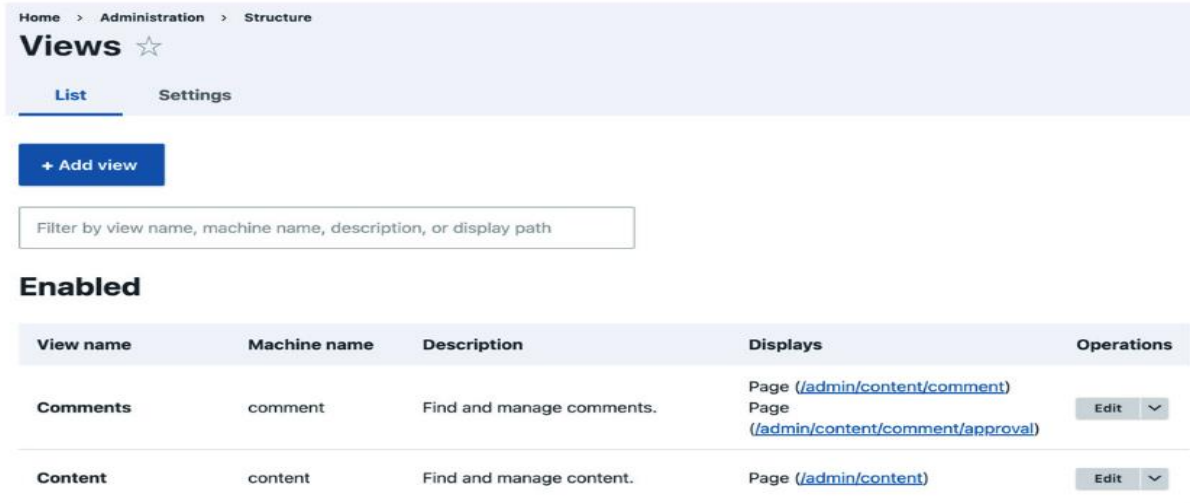


# Content Management Systems

## Creating a blog landing page

How to do it...

1. Go to Structure and then Views. This will bring you to the administrative overview of all the views created. Click on Add view to create a new view



The screenshot shows the Drupal Views administrative interface. At the top, there is a breadcrumb trail: Home > Administration > Structure. Below this, the 'Views' section is highlighted with a star icon. There are two tabs: 'List' (active) and 'Settings'. A blue button labeled '+ Add view' is prominently displayed. Below the button is a search box with the placeholder text 'Filter by view name, machine name, description, or display path'. The main content area is titled 'Enabled' and contains a table listing the available views.

| View name | Machine name | Description               | Displays                                                                                                                                                       | Operations |
|-----------|--------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Comments  | comment      | Find and manage comments. | Page ( <a href="/admin/content/comment">/admin/content/comment</a> )<br>Page ( <a href="/admin/content/comment/approval">/admin/content/comment/approval</a> ) | Edit ▾     |
| Content   | content      | Find and manage content.  | Page ( <a href="/admin/content">/admin/content</a> )                                                                                                           | Edit ▾     |

Figure 3.1 – The Views listing overview

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## Creating a blog landing page

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2. The first step is to provide the view name of Blog, which will serve as the administrative and (by default) displayed title.
3. Next, we will modify the View settings section. We want to display Content of the Article type and leave the tagged with field empty. This will force the view to only show content of the Article content type.
4. Check the Create a page option. The Page title and Path fields will be automatically populated based on the view name and can be modified as desired. For now, leave the display and other settings at their default values:

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## Creating a blog landing page

**View basic information**

**View name \***  
 Machine name: blog [\[Edit\]](#)

☐ Description

**View settings**

Show:  of type:  tagged with:  sorted by:

**Page settings**

☒ Create a page

Page title

Path

**Page display settings**

Display format:  of:

Figure 3.2 – Overview of the view creation form

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## Creating a blog landing page

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5. Click on Save and edit to continue modifying your new view.
6. In the middle column, under the Page Settings section, we will change the Menu item settings. Click on No menu to change the menu option.
7. Select Normal menu entry. Provide a menu link title and an optional description. Set Parent to <Main Navigation>

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## Creating a blog landing page

**Page: Menu link entry** ×

**Type**  
☐ No menu entry  
☒ Normal menu entry  
☐ Menu tab  
☐ Default menu tab

**Menu link title**

**Description**  
  
Shown when hovering over the menu link.  
☐ Show as expanded  
If selected and this menu link has children, the menu will always appear expanded.

**Parent**  
  
The maximum depth for a link and all its children is fixed. Some menu links may not be available as parents if selecting them would exceed this limit.

**Weight**  
  
In the menu, the heavier links will sink and the lighter links will be positioned nearer the top.

Apply

Cancel

Figure 3.3 – The menu settings form for the view

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## Creating a blog landing page

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8. Click on Apply at the bottom of the form in the dialog.
9. Click on Save to save your view.
10. Once your view has been saved, click on Back to site from the administrative menu. You will now see the link in your Drupal site's main menu.

# Content Management Systems

## Creating a blog landing page

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How it works...

The first step for creating a view involves selecting the type of data you will be displaying. This is referred to as the base table, which can be any type of entity or data specifically exposed to Views.

### **Content and nodes**

Nodes are labeled as Content in Views, and you will find this interchanged terminology throughout Drupal.

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## Creating a blog landing page

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When creating a Views page, we add a menu path that can be accessed. It tells Drupal to invoke Views to render the page, which will load the view you created and render it.

Then, the Views module registers that path as a route in Drupal's routing system to a controller provided by itself, which renders the view display.

There are display style and row plugins that format the data to be rendered.



The views module is one of the most versatile and used modules in Drupal core.

### Views and displays

When working with views, you will see some different terminologies.

A view contain multiple displays. Each display is of a certain type. The view modules comes with following display types:

**Attachment:** This is a display that becomes attached to another display in the same view.

**Block:** This allows you to place the view as a block on your Drupal site.

**Embed:** The display is meant to be embedded programmatically.

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## Views and displays

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**Entity Reference:** This allows Views to provide results for an entity reference field

**Feed:** This display returns an XML-based feed that can be attached to another display to render a feed icon so that users can subscribe to the content.

**Page:** This allows you to display the view from a specific path.

Each display has its own configuration.

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Create a block of recent blogs

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# Content Management Systems

## Create a block of recent blogs

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- A view can contain multiple displays. Each display inherits defaults, such as its style and row format, filters, sorts, pager, and more.
- Each display type might have unique settings, such as the page settings where the menu link was configured in the previous steps.
- In this recipe, we will add a block display so that we can list the five most recent articles by title anywhere on the site.

# Content Management Systems

## Create a block of recent blogs

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How to do it...

- 1. Go to Structure and then Views. This will bring you to the administrative overview of all the views that are created.
- 2. Find the Blog view that was created in the previous steps and click on Edit.
- 3. Under Displays, where it says Page, click on the Add button and select Block from the drop- down menu. This will create a new block display, which we can configure:

# Content Management Systems

## Create a block of recent blogs

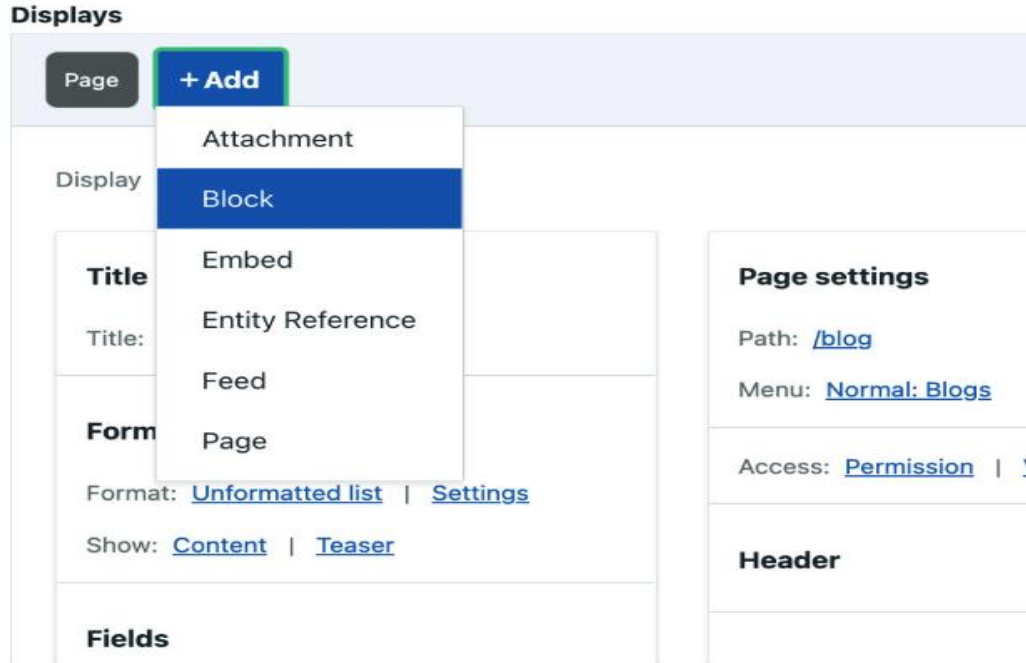


Figure 3.4 – The menu for adding a new display

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## Create a block of recent blogs

---

4. In the Format section, click on Content (next to Show) to configure the row format used.
5. In the dialog that opens, we need to make sure we only modify the row format for this display. In the For drop-down menu, select This block (override) so that the changes are only for this display.
6. Select the Fields option from the radio buttons and click on Apply (this display) to set the row format to use individual fields instead of display modes:

# Content Management Systems

## Create a block of recent blogs

**Block: How should each row in this view be styled** ×

**For**  

This block (override) ▾

☐

 Content

☒

 Fields

☐

 Search results

You may also adjust the [settings](#) for the currently selected row style.

Apply (this display)

Cancel

Figure 3.5 – The style format form dialog



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## Create a block of recent blogs

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7. A new dialog will appear that allows you to configure options for this row format. Use the defaults and click on Apply.

### **Important note**

Changing to using the Fields row format will automatically add the Title field, which has been configured to link to the piece of content.

8. We only want this block to display the five most recent blogs. Under the Pager section, click on the field next to Use Pager to launch the dialog for configuring the pager.

9. Just like the row format, select This block (override) from the For select list. Then, choose the Display a specified number of items option. Click on Apply (this display):

# Content Management Systems

## Create a block of recent blogs

**Block: Select pager** ×

**For**  

This block (override) ▾

☒

 Display a specified number of items

☐

 Display all items☐☐

Apply (this display)

Cancel

Figure 3.6 – The pager selection form dialog

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## Create a block of recent blogs

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10. In the next dialog, which allows you to configure the pager settings, change Items to display to 5. Click on Apply to finish changing the pager.
11. Click on Save to save your changes to the view and create the block display.
12. Go to Structure and Block layout to place the block on your Drupal site. Click on Place block for the Sidebar first region.
13. Filter the list by typing in your view's name: Blog. Click on Place block to add your view's block to the block layout.

# Content Management Systems

## Create a block of recent blogs

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14. We do not want this block to display on our blog landing page. In the Visibility section, add / blog to the Pages textbox. Change the radio option to Hide for the listed pages.
15. This prevents the block from being displayed in the /blog path, which is our view for the list of articles.
16. Finally, click on the Save block to commit your changes.

# Content Management Systems

## Create a block of recent blogs

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### How it works...

Blocks, in Drupal, are a type of plugin. These blocks can be embedded into the site layout and can be shown or hidden based on a variety of visibility settings.

The Views module integrates with the block module to allow view displays to be placed as blocks. This empowers Drupal site builders to create dynamic content displays that can be used throughout a site.

# Content Management Systems

## Create a block of recent blogs

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We used Layout Builder to place blocks that contained field values for content. You could place any kind of block using Layout Builder, including block displays in your view. Content creators can create landing pages using dynamic content provided by views

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# Content Management Systems

## Create a block of recent blogs

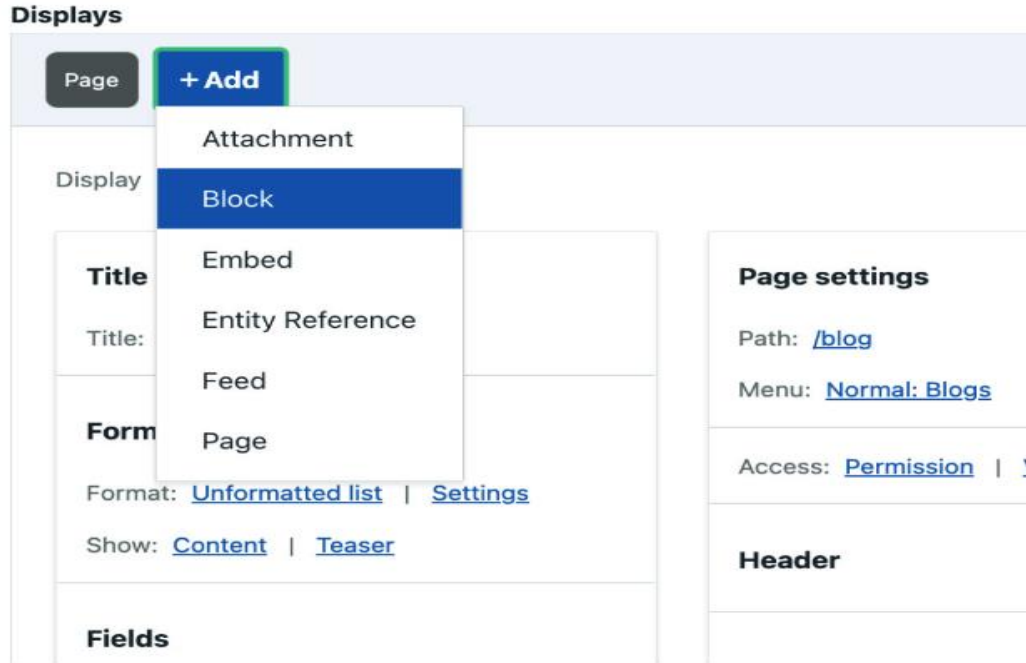


Figure 3.4 – The menu for adding a new display

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# Content Management Systems

## Create a block of recent blogs

**Block: How should each row in this view be styled** ×

**For**  

This block (override) ▾

☐ Content

☒ Fields

☐ Search results

You may also adjust the [settings](#) for the currently selected row style.

Apply (this display)

Cancel

Figure 3.5 – The style format form dialog

# Content Management Systems

## Create a block of recent blogs

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7. A new dialog will appear that allows you to configure options for this row format. Use the defaults and click on Apply.

### **Important note**

Changing to using the Fields row format will automatically add the Title field, which has been configured to link to the piece of content.

8. We only want this block to display the five most recent blogs. Under the Pager section, click on the field next to Use Pager to launch the dialog for configuring the pager.

9. Just like the row format, select This block (override) from the For select list. Then, choose the Display a specified number of items option. Click on Apply (this display):

# Content Management Systems

## Create a block of recent blogs

**Block: Select pager** ×

**For**  

This block (override) ▾

☒

 Display a specified number of items

☐ Display all items

☐ Paged output, full pager

☐ Paged output, mini pager

You may also adjust the [settings](#) for the currently selected pager.

Apply (this display)

Cancel

Figure 3.6 – The pager selection form dialog

# Content Management Systems

## Create a block of recent blogs

---

10. In the next dialog, which allows you to configure the pager settings, change Items to display to 5. Click on Apply to finish changing the pager.
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## Create a block of recent blogs

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We used Layout Builder to place blocks that contained field values for content. You could place any kind of block using Layout Builder, including block displays in your view. Content creators can create landing pages using dynamic content provided by views

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# Content Management Systems

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**Exposing filters and sorts to users to control listings**

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# Content Management Systems

## Exposing filters and sorts to users to control listings

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- The Views module supports exposing filters and sorts on a view display for users to interact with to adjust the results.
- This can be done to allow users to filter by text or adjust the ordering of the results.
- In this step, we will modify the view used to create a blog page.

# Content Management Systems

## Exposing filters and sorts to users to control listings

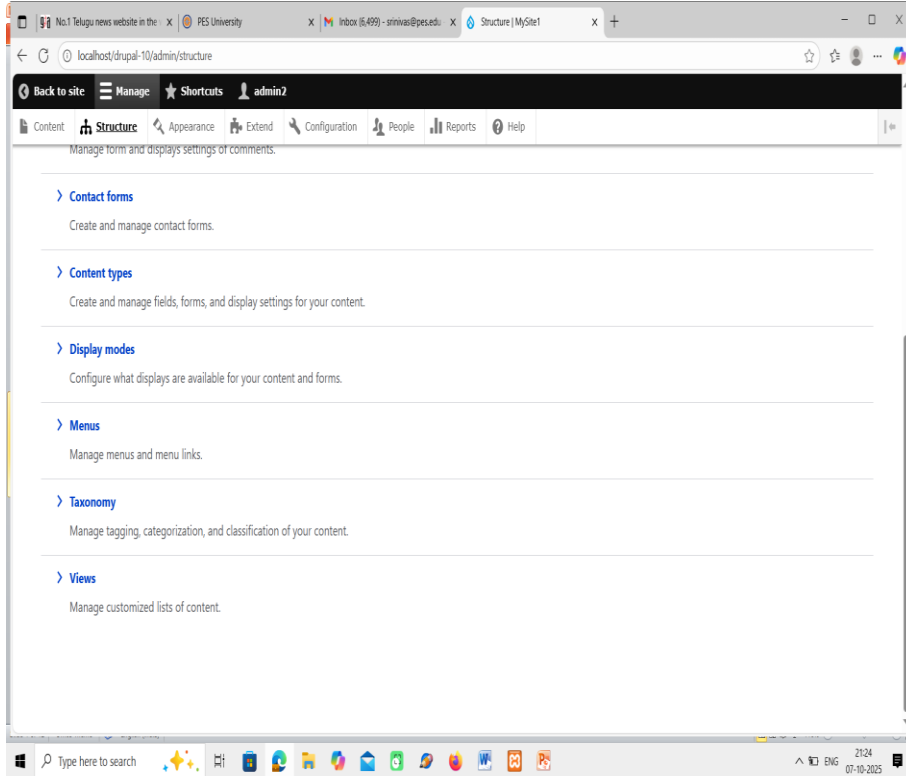
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How to do it...

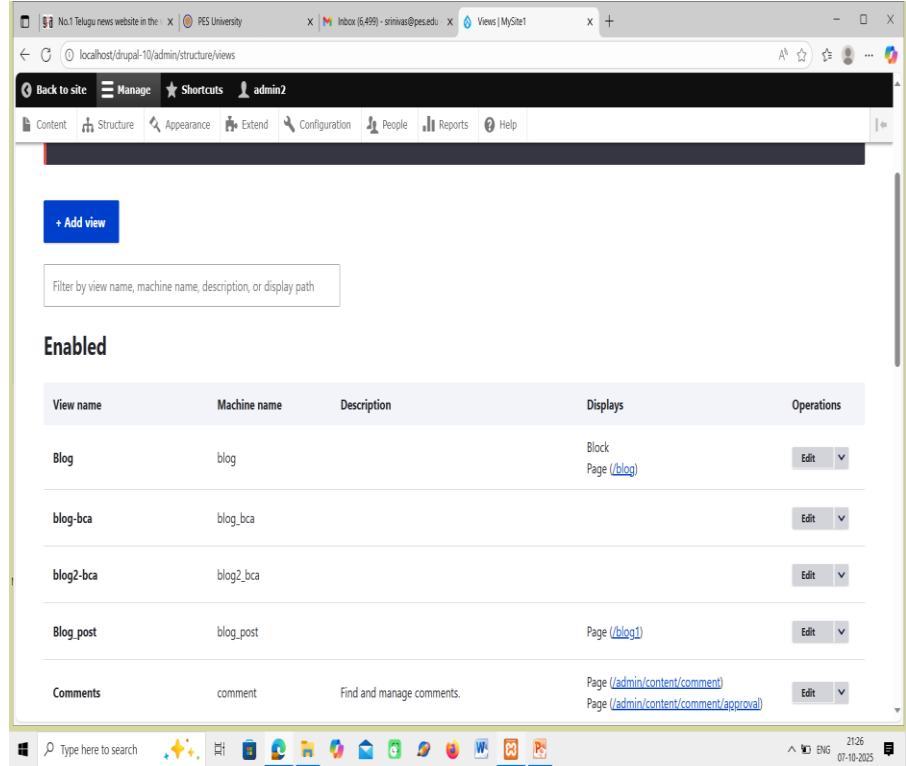
- 1. Go to Structure and then Views. This will bring you to an administrative overview of all the views that have been created.
- 2. Find the Blog view that was created in the first step and click on Edit.
- 3. Click on Add in the Filter criteria section of the form.

# Content Management Systems

## Exposing filters and sorts to users to control listings



The screenshot shows the Drupal 10 admin interface at the URL `localhost:drupal-10/admin/structure`. The breadcrumb trail is `Back to site > Manage > Shortcuts > admin2`. The main navigation bar includes links for Content, Structure, Appearance, Extend, Configuration, People, Reports, and Help. The Structure page title is "Manage form and displays settings of comments." The page lists several configuration sections: Contact forms, Content types, Display modes, Menus, Taxonomy, and Views. Each section has a brief description of its function.

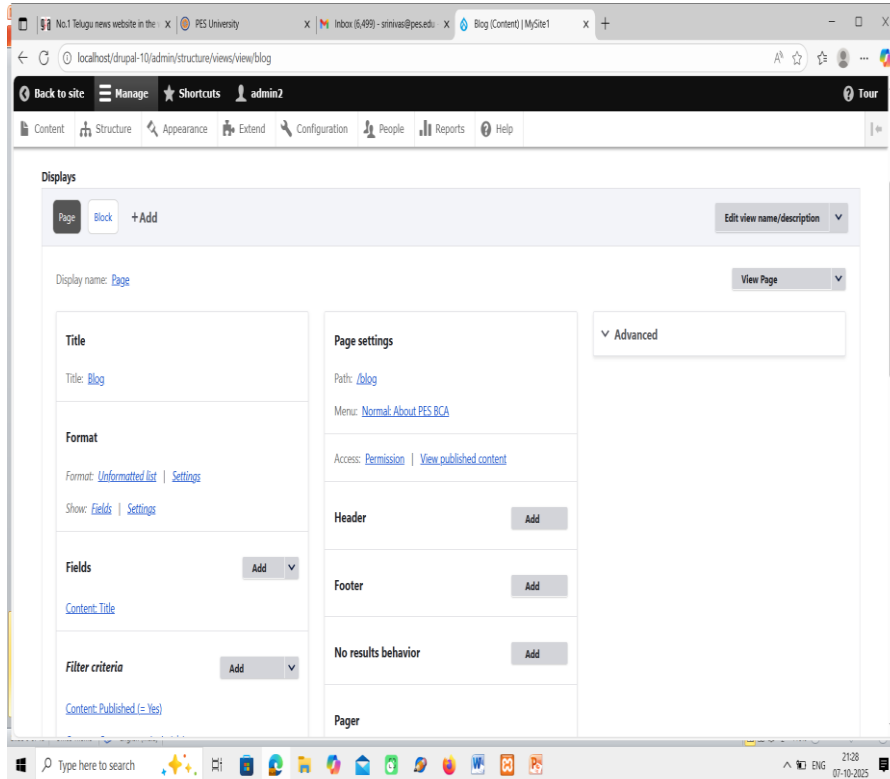


The screenshot shows the Drupal 10 admin interface at the URL `localhost:drupal-10/admin/structure/views`. The breadcrumb trail is `Back to site > Manage > Shortcuts > admin2`. The main navigation bar is the same as the previous screenshot. The Views page title is "Manage form and displays settings of comments." There is a blue "+ Add view" button and a search box labeled "Filter by view name, machine name, description, or display path". Below this is a table titled "Enabled" showing a list of views.

| View name | Machine name | Description               | Displays                                                                                                    | Operations |
|-----------|--------------|---------------------------|-------------------------------------------------------------------------------------------------------------|------------|
| Blog      | blog         |                           | Block<br>Page ( <a href="#">/blog</a> )                                                                     | Edit ▼     |
| blog-bca  | blog_bca     |                           |                                                                                                             | Edit ▼     |
| blog2-bca | blog2_bca    |                           |                                                                                                             | Edit ▼     |
| Blog_post | blog_post    |                           | Page ( <a href="#">/blog1</a> )                                                                             | Edit ▼     |
| Comments  | comment      | Find and manage comments. | Page ( <a href="#">/admin/content/comment</a> )<br>Page ( <a href="#">/admin/content/comment/approval</a> ) | Edit ▼     |

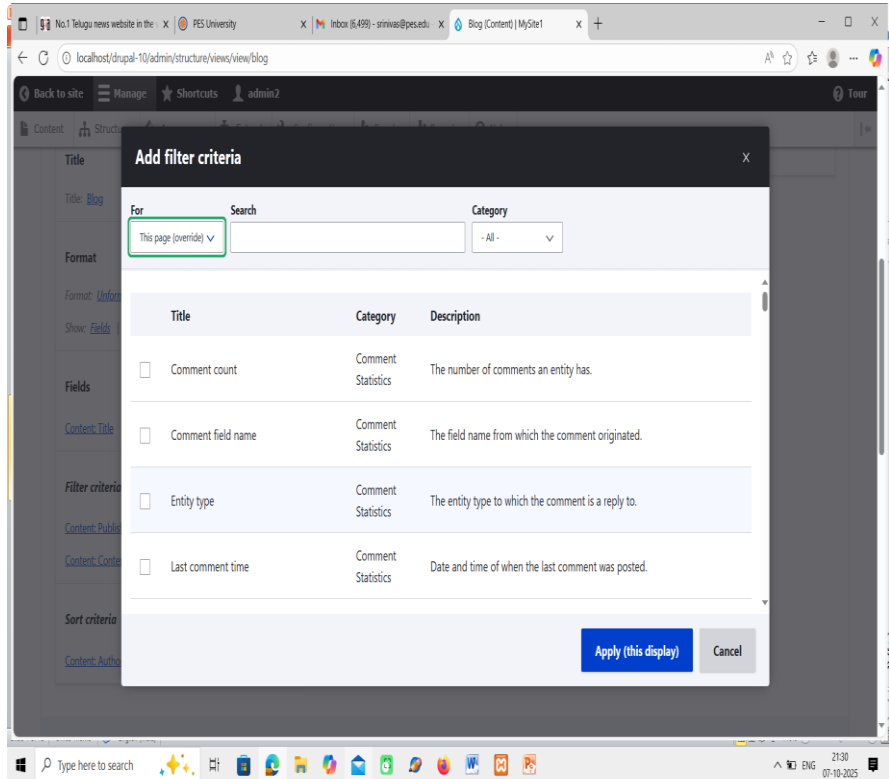
# Content Management Systems

## Exposing filters and sorts to users to control listings



The screenshot shows the Drupal 10 admin interface for configuring a 'Blog' view. The 'Displays' section is active, showing the 'Page' display type. The configuration is divided into several sections:

- Title:** The title is 'Blog'.
- Format:** The format is 'Unformatted list'.
- Fields:** The 'Content: Title' field is added.
- Filter criteria:** The 'Content: Published (= Yes)' filter is added.
- Page settings:** The path is '/blog', the menu is 'Normal: About PES BCA', and the access is 'Permission | View published content'.
- Header, Footer, No results behavior, and Pager:** Each of these sections has an 'Add' button to configure further options.



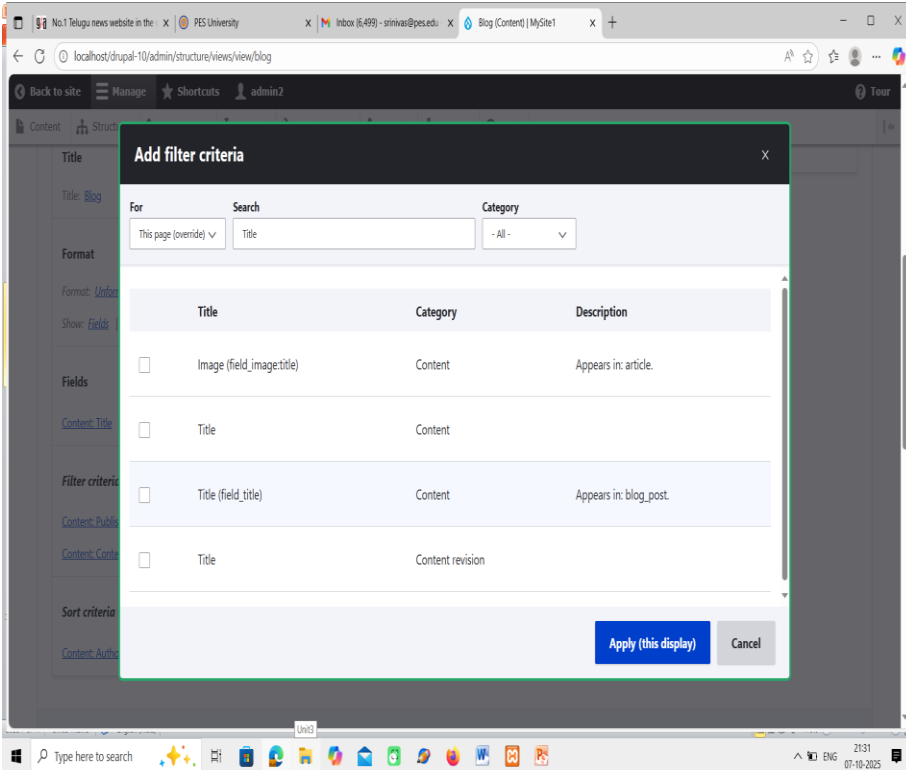
The screenshot shows the 'Add filter criteria' modal in the Drupal 10 admin interface. The modal is titled 'Add filter criteria' and has a close button (X) in the top right corner. It contains the following fields and options:

- For:** A dropdown menu with 'This page (override)' selected.
- Search:** An empty text input field.
- Category:** A dropdown menu with 'All' selected.
- Table of filter criteria:** A table with columns 'Title', 'Category', and 'Description'. It lists several filter criteria with checkboxes for selection.
- Buttons:** 'Apply (this display)' and 'Cancel' buttons at the bottom right.

| Title                                       | Category           | Description                                         |
|---------------------------------------------|--------------------|-----------------------------------------------------|
| <input type="checkbox"/> Comment count      | Comment Statistics | The number of comments an entity has.               |
| <input type="checkbox"/> Comment field name | Comment Statistics | The field name from which the comment originated.   |
| <input type="checkbox"/> Entity type        | Comment Statistics | The entity type to which the comment is a reply to. |
| <input type="checkbox"/> Last comment time  | Comment Statistics | Date and time of when the last comment was posted.  |

# Content Management Systems

## Exposing filters and sorts to users to control listings



The screenshot shows the Drupal 10 admin interface for a 'Blog' view. A modal dialog titled 'Add filter criteria' is open, allowing the user to select filter criteria for the view. The dialog includes a 'For' dropdown set to 'This page (override)', a 'Search' input field containing 'Title', and a 'Category' dropdown set to '- All -'. Below these fields is a table of available filter criteria:

|                          | Title                     | Category         | Description            |
|--------------------------|---------------------------|------------------|------------------------|
| <input type="checkbox"/> | Image (field_image:title) | Content          | Appears in: article.   |
| <input type="checkbox"/> | Title                     | Content          |                        |
| <input type="checkbox"/> | Title (field_title)       | Content          | Appears in: blog_post. |
| <input type="checkbox"/> | Title                     | Content revision |                        |

At the bottom of the dialog are two buttons: 'Apply (this display)' and 'Cancel'.



# Content Management Systems

## Exposing filters and sorts to users to control listings

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- 4. In the dialog that opens, we need to make sure we only modify the row format for this display. In the For drop-down menu, select This page (override) so that the changes are only for this page display and not the block display created in the previous step.
- 5. Select Title in the Content category and click on Add (this display):

# Content Management Systems

## Exposing filters and sorts to users to control listings

**Add filter criteria** ×

**For**  
This page (override) ▼

**Search**  
Title

**Category**  
- All - ▼

|                                     | Title                     | Category         | Description          |
|-------------------------------------|---------------------------|------------------|----------------------|
| <input type="checkbox"/>            | Image (field_image:title) | Content          | Appears in: article. |
| <input checked="" type="checkbox"/> | Title                     | Content          |                      |
| <input type="checkbox"/>            | Title                     | Content revision |                      |

Selected: Title

Apply (this display)

Cancel

Figure 3.7 – The add filter form dialog

# Content Management Systems

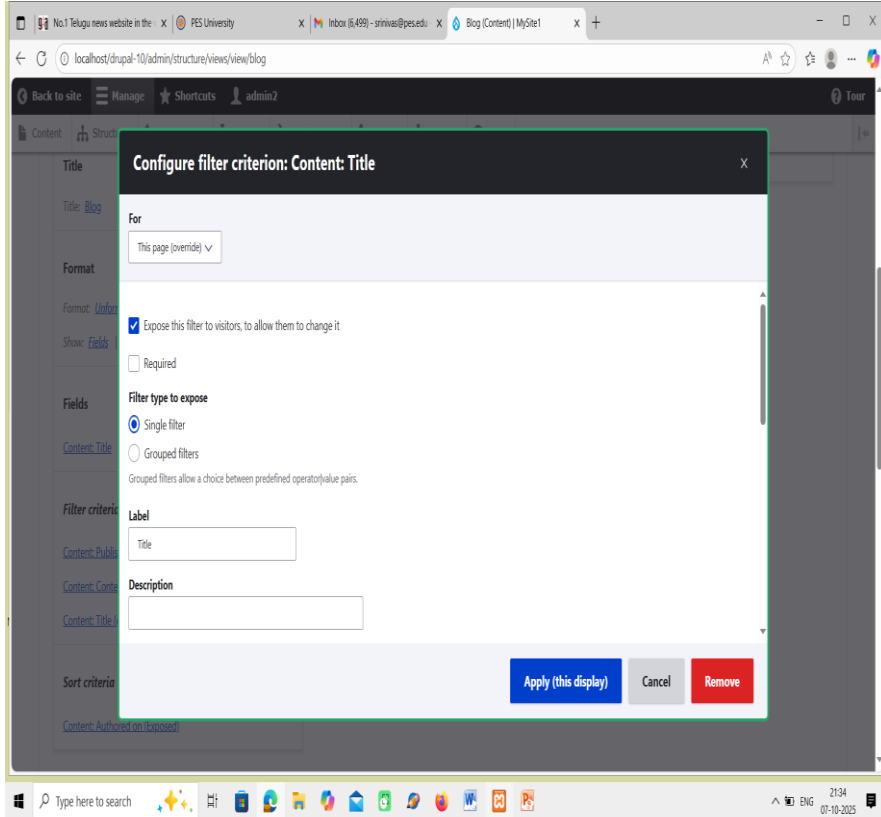
## Exposing filters and sorts to users to control listings

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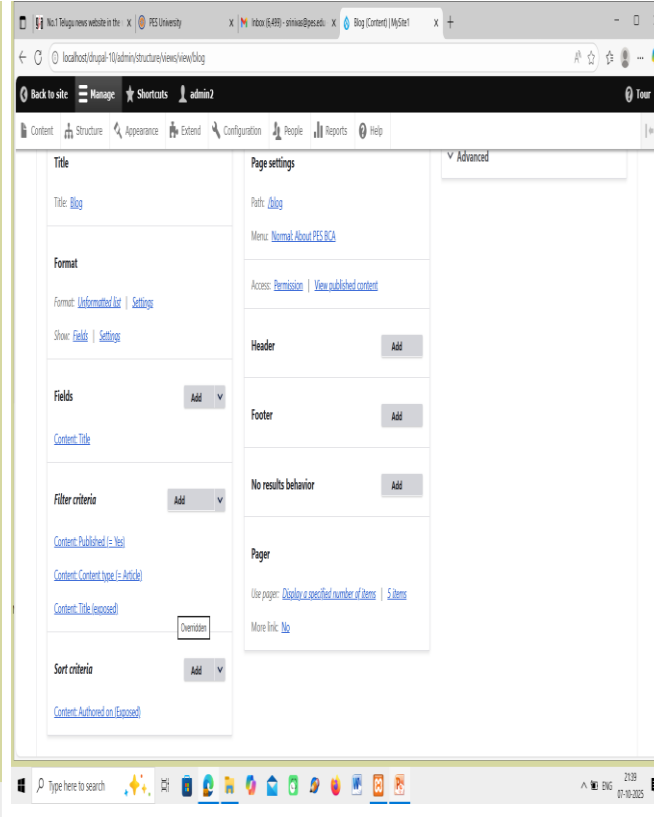
6. Check the Expose this filter to visitors, to allow them to change it checkbox to make this an exposed filter. In the exposed filter configuration form, change Operator from Is equal to To Contains. This will allow for more flexible searching. Click on Apply (this display) to add the new filter.

# Content Management Systems

## Exposing filters and sorts to users to control listings



The screenshot shows the Drupal admin interface with a modal dialog titled "Configure filter criterion: Content: Title". The dialog is used to configure how the "Content: Title" filter is exposed to visitors. It includes a "For" dropdown set to "This page (override)", a checked option "Expose this filter to visitors, to allow them to change it", and a "Filter type to expose" section with "Single filter" selected. The "Label" field is set to "Title" and the "Description" field is empty. At the bottom, there are buttons for "Apply (this display)", "Cancel", and "Remove".



The screenshot shows the Drupal admin interface for a content type's "Page settings". The "Title" field is set to "Blog". The "Format" is set to "Unformatted text" with a link to "Settings". The "Fields" section shows "Content: Title" with an "Add" button. The "Filter criteria" section shows "Content: Published (= Yes)", "Content: Content type (= Article)", and "Content: Title (exposed)" with an "Override" button. The "Sort criteria" section shows "Content: Authored on (exposed)" with an "Add" button. The "Page settings" section includes "Path" (Blog), "Menu" (Normal: About PES UCA), "Access" (Permission), "Header" (Add), "Footer" (Add), "No results behavior" (Add), and "Pager" (Use pager: Display a specified number of items | 5 items, More link: No).

# Content Management Systems

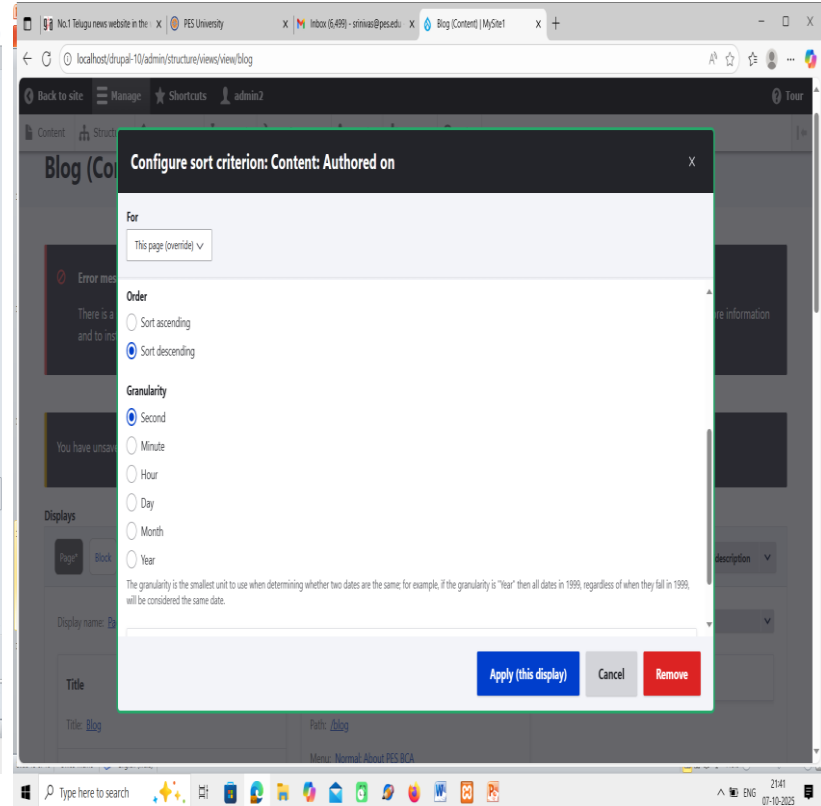
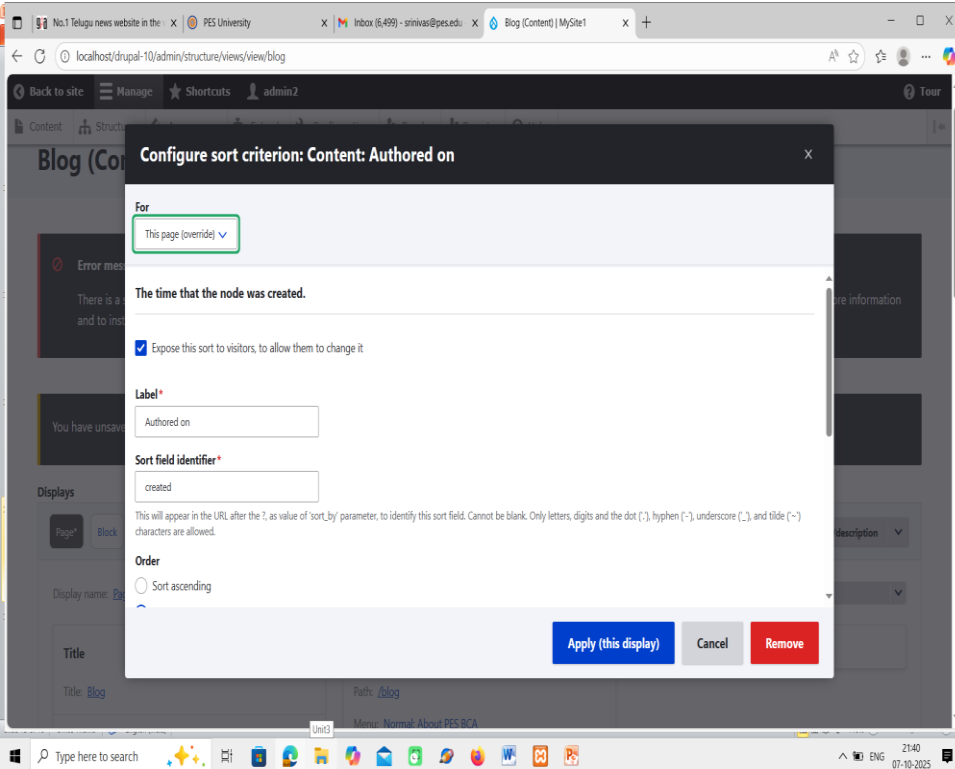
## Exposing filters and sorts to users to control listings

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7. In the Sort criteria section, click on Content: Authored on (desc). Select This page (override) from the For select list.
8. Check the Expose this sort to visitors, to allow them to change it checkbox to make this an exposed sort.
9. Click on Apply (this display) to update the sorting configuration.
10. Click on Save to save your changes to the view.
11. Now, when you view /blog, you can search the articles and change their sort order

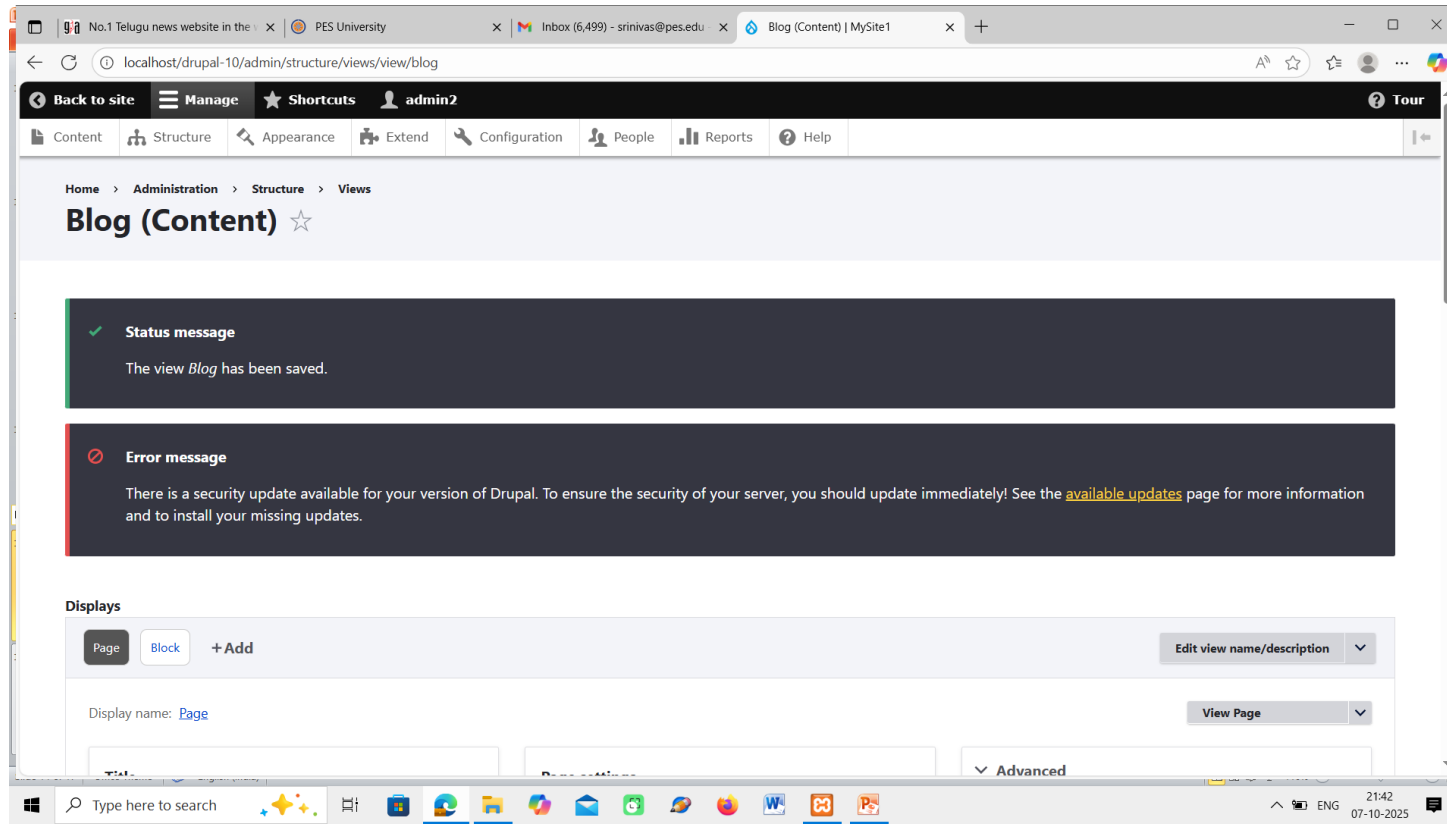
# Content Management Systems

## Exposing filters and sorts to users to control listings



# Content Management Systems

## Exposing filters and sorts to users to control listings



The screenshot displays the Drupal 10 administrative interface for configuring a content view. The browser's address bar shows the URL `localhost/drupal-10/admin/structure/views/view/blog`. The top navigation bar includes links for 'Back to site', 'Manage', 'Shortcuts', and the user 'admin2'. A secondary menu contains 'Content', 'Structure', 'Appearance', 'Extend', 'Configuration', 'People', 'Reports', and 'Help'. The breadcrumb trail is 'Home > Administration > Structure > Views'. The main heading is 'Blog (Content)' with a star icon. Below this, there are two messages: a green 'Status message' stating 'The view *Blog* has been saved.' and a red 'Error message' about a security update. The 'Displays' section shows a 'Page' display selected, with buttons for 'Block' and '+ Add'. An 'Edit view name/description' dropdown is also present. At the bottom, the 'Display name' is set to 'Page' and a 'View Page' button is visible. The system tray at the bottom of the screen shows the time as 21:42 on 07-10-2025.

# Content Management Systems

## Exposing filters and sorts to users to control listings



The screenshot shows the top of a Drupal website. A blue header bar contains the text "Drupal Development Cookbook". To the right of the header are links for "Blogs", "Home", a search icon, "My account", and "Log out". Below the header, a link for "Home" is visible. The main heading is "Blog" followed by a star icon. Below the heading are three filter boxes: "Title" with an empty text input, "Sort by" with a dropdown menu showing "Authored on", and "Order" with a dropdown menu showing "Desc". To the right of these filters is a blue "Apply" button.

Drupal Development Cookbook

Blogs Home 🔍 My account Log out

Home

## Blog ☆

Title Sort by Order

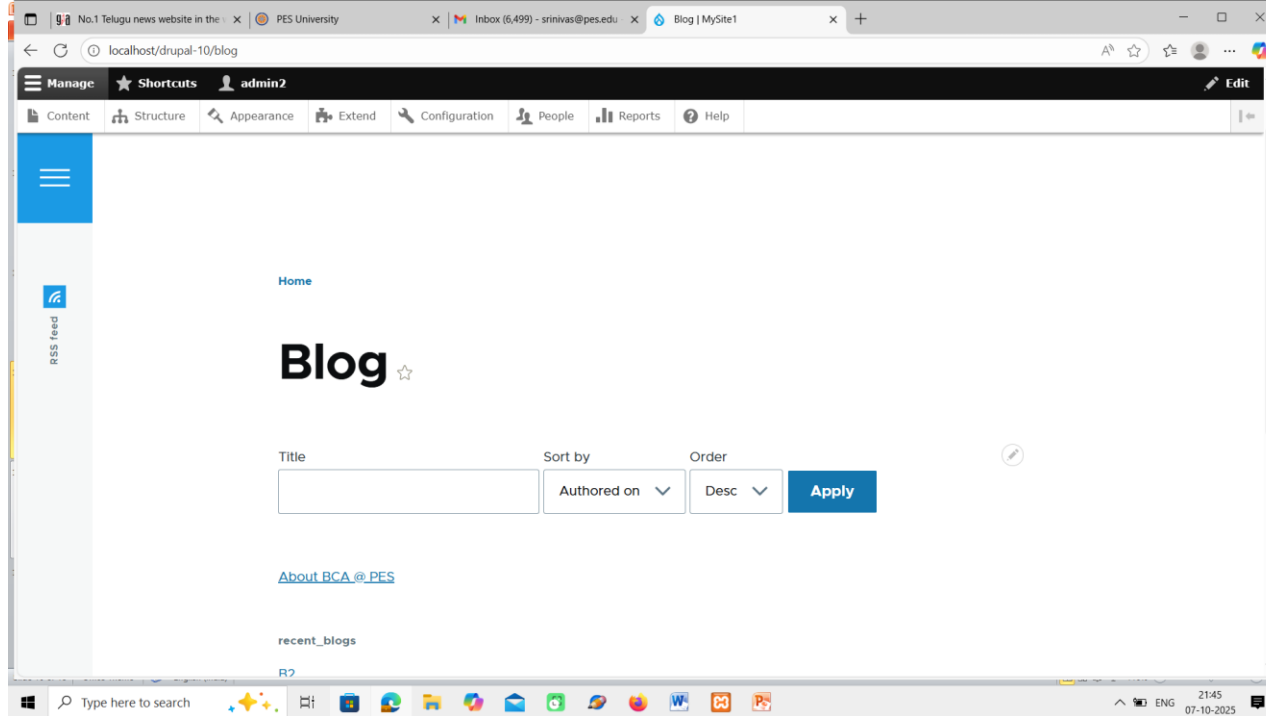
Authored on ▼ Desc ▼ [Apply](#)

Figure 3.8 – The Blog landing page with exposed filters and sorts



# Content Management Systems

## Exposing filters and sorts to users to control listings



The screenshot shows a web browser window displaying a Drupal 10 blog page. The browser's address bar shows the URL `localhost/drupal-10/blog`. The page features a top navigation bar with links for **Manage**, **Shortcuts**, and **admin2**. Below this is a secondary navigation bar with icons for **Content**, **Structure**, **Appearance**, **Extend**, **Configuration**, **People**, **Reports**, and **Help**. On the left side, there is a blue sidebar with a menu icon and an **RSS feed** icon. The main content area is titled **Home** and **Blog** (with a star icon). Below the title, there are three input fields: **Title** (with an empty text box), **Sort by** (with a dropdown menu showing **Authoried on**), and **Order** (with a dropdown menu showing **Desc**). To the right of these fields is a blue **Apply** button. Below the filters, there is a link [About BCA @ PES](#) and a section labeled **recent\_blogs**. The bottom of the page shows a Windows taskbar with various application icons and a system clock indicating 21:45 on 07-10-2025.

# Content Management Systems

## Exposing filters and sorts to users to control listings

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How it works...

When a filter or sort is exposed, the Views module attaches a form to the view display. This form is controlled by the `\Drupal\views\Form\ViewsExposedForm` class.

This reads query parameter values from the URL and maps them to known exposed filters and sorts. It applies the values to the filter and sort handlers, overriding their default values so that the query is adjusted.

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# Content Management Systems

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## Adding related data in view with relationships

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# Content Management Systems

## Adding related data in view with relationships

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Views is a visual query builder. When you first create a view, a base table is specified from which to pull data. The Views module automatically knows how to join tables for field data, such as body text or custom-attached fields.

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## Adding related data in view with relationships

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When using an entity reference field, you can display the value as the identifier, the referenced entity's label, or the entire rendered entity. However, if you add a relationship based on a reference field, you will have access to display any of that entity's available fields.

- In this step, we will update the Files view, used for administering files, to display the username of the user who uploaded the file.

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## Adding related data in view with relationships

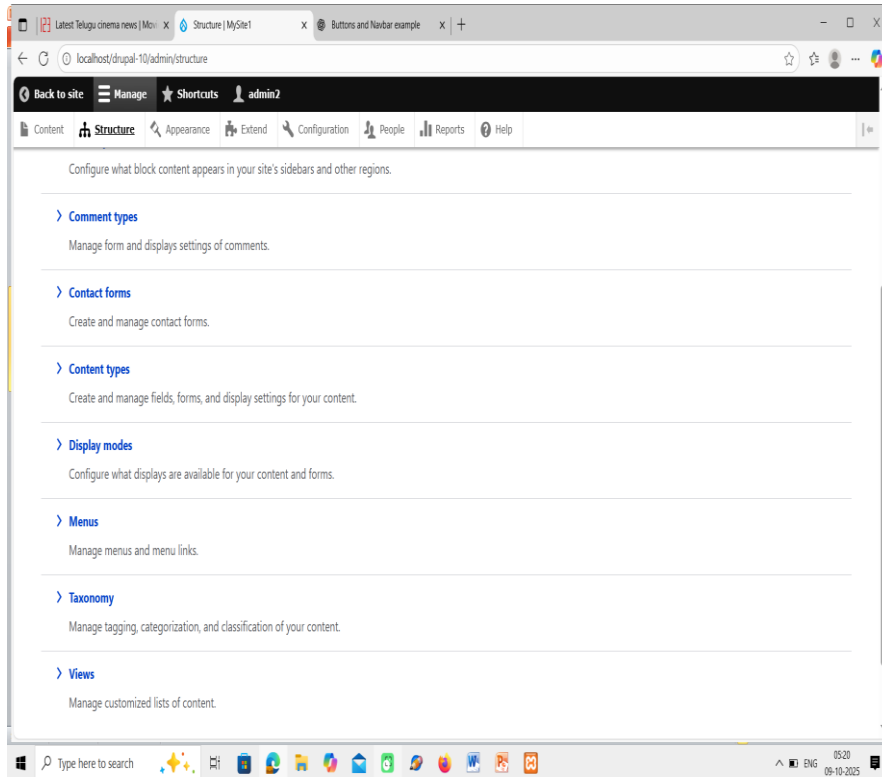
---

How to do it...

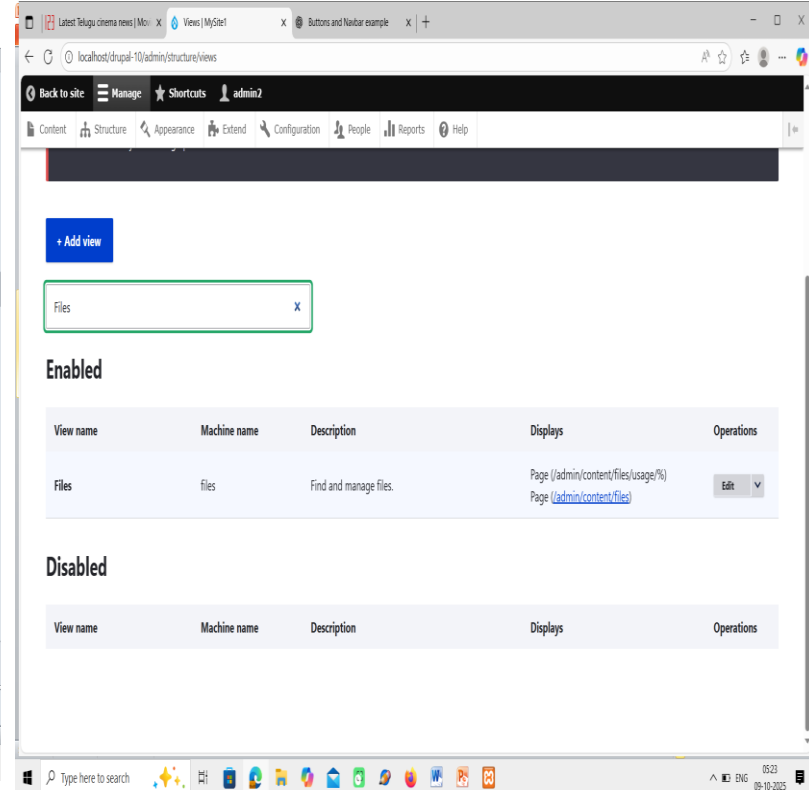
1. Go to Structure and then Views. This will bring you to the administrative overview of all the views that have been created.
2. Find the Files view and click on Edit:

# Content Management Systems

## Adding related data in view with relationships



The screenshot shows the Drupal 10 admin interface at the `admin/structure` path. The left sidebar contains a list of configuration categories: Comment types, Contact forms, Content types, Display modes, Menus, Taxonomy, and Views. The 'Views' category is currently selected, showing the text 'Manage customized lists of content.'



The screenshot shows the Drupal 10 admin interface at the `admin/structure/views` path. A '+ Add view' button is visible, followed by a search box containing the text 'Files'. Below this, the 'Enabled' section contains a table with the following data:

| View name | Machine name | Description            | Displays                                                           | Operations           |
|-----------|--------------|------------------------|--------------------------------------------------------------------|----------------------|
| Files     | files        | Find and manage files. | Page (/admin/content/files/usage/%)<br>Page (/admin/content/files) | <a href="#">Edit</a> |

Below the 'Enabled' section, the 'Disabled' section is visible, which is currently empty.

# Content Management Systems

## Adding related data in view with relationships

|                      |               |                                         |                                                                                                                                                         |        |
|----------------------|---------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Custom block library | block_content | Find and manage custom blocks.          | Page<br>( <a href="/admin/structure/block/block-content">/admin/structure/block/block-content</a> )                                                     | Edit ▾ |
| Files                | files         | Find and manage files.                  | Page<br>( <a href="/admin/content/files/usage/%">/admin/content/files/usage/%</a> )<br>Page ( <a href="/admin/content/files">/admin/content/files</a> ) | Edit ▾ |
| Frontpage            | frontpage     | All content promoted to the front page. | Feed ( <a href="/rss.xml">/rss.xml</a> )<br>Page ( <a href="/node">/node</a> )                                                                          | Edit ▾ |
| My Content           | my_content    |                                         | Page ( <a href="/user/%user/content">/user/%user/content</a> )                                                                                          | Edit ▾ |

Figure 3.12 – The Files view in the Views listing



# Content Management Systems

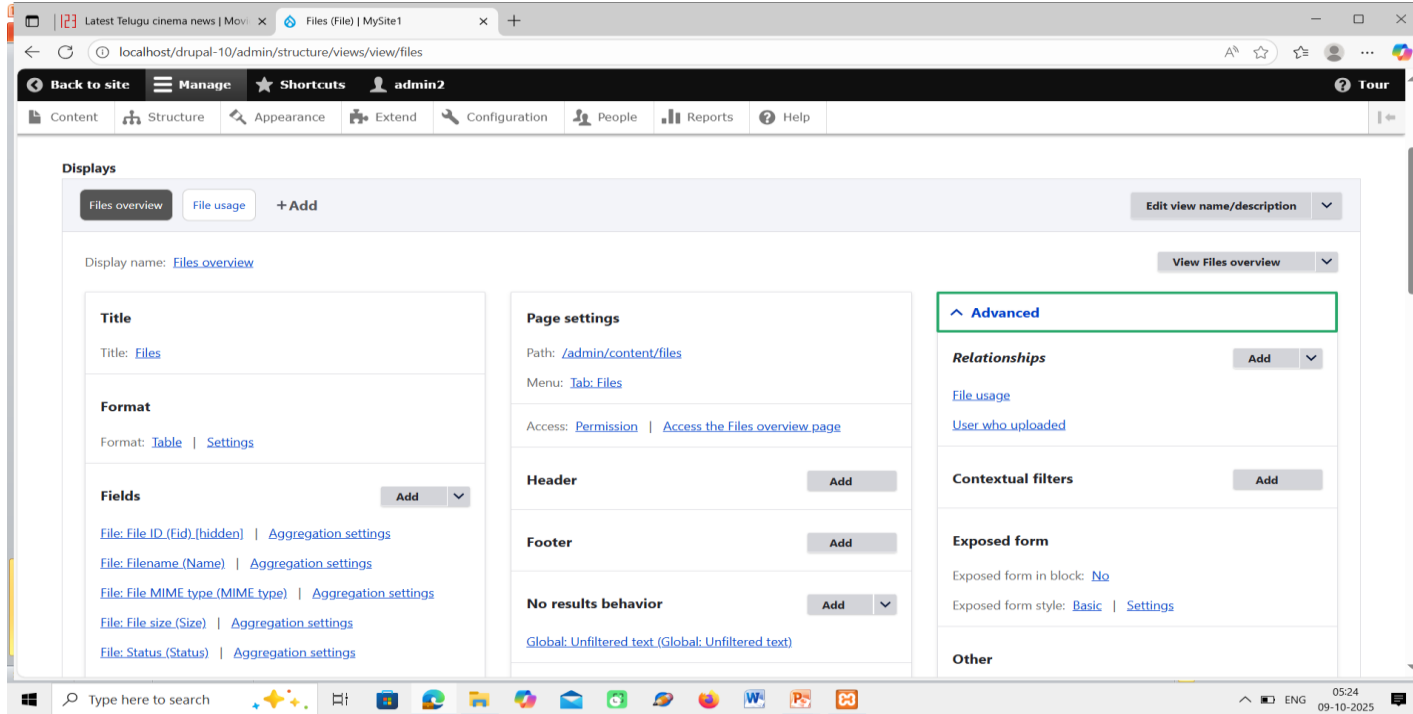
## Adding related data in view with relationships

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3. Click on Advanced to expand the section and then click on Add, which is next to Relationships.
4. Search for User. Select the User who uploaded and click on Apply (this display).
5. Next, we will be presented with a configuration form for the relationship. Click on Apply (this display) to use the defaults.

# Content Management Systems

## Adding related data in view with relationships

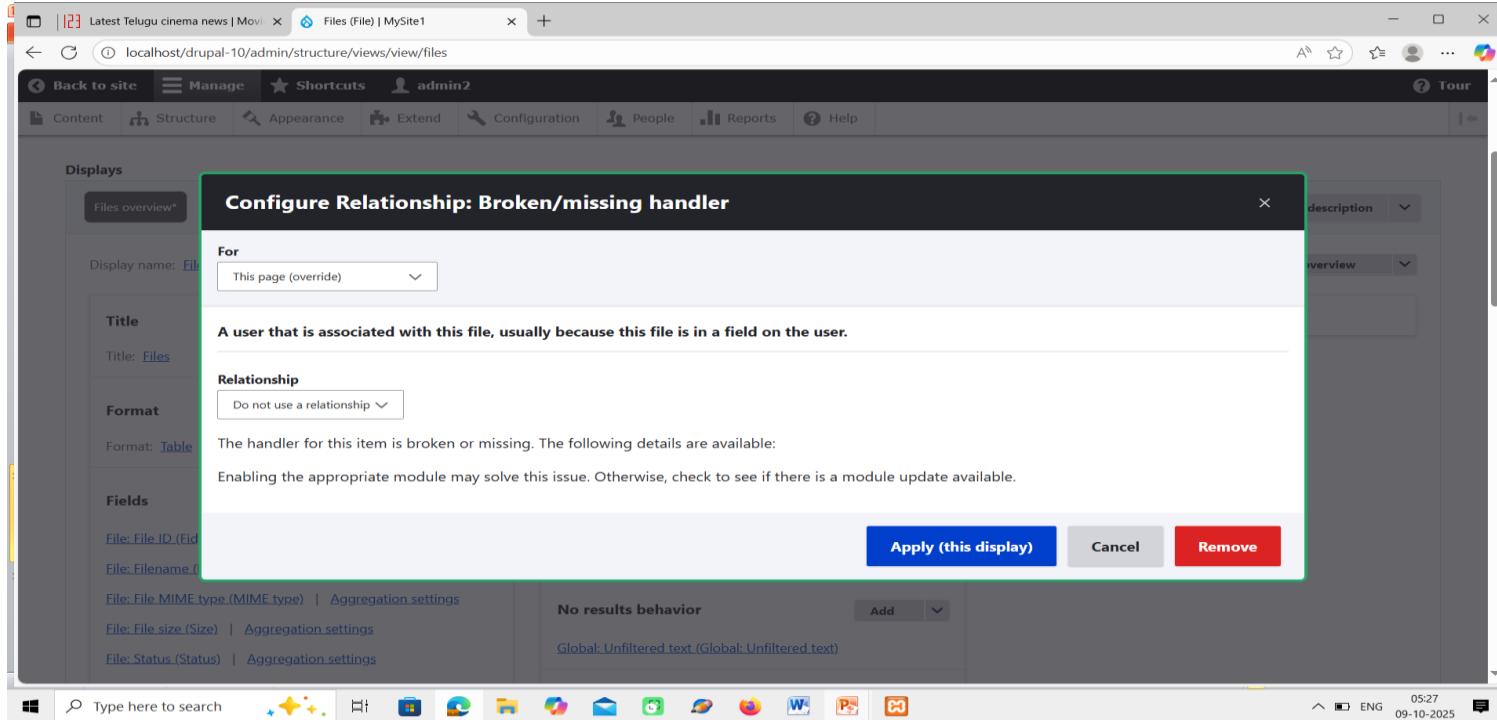


The screenshot displays the Drupal 10 administrative interface for configuring a view named 'Files overview'. The interface is divided into several sections for configuration:

- Displays:** Includes tabs for 'Files overview' and 'File usage', with an '+ Add' button to create new displays.
- Title:** The title is set to 'Files'.
- Format:** The format is set to 'Table'.
- Fields:** A list of available fields for the table view, including 'File: File ID (Fid) [hidden]', 'File: Filename (Name)', 'File: File MIME type (MIME type)', 'File: File size (Size)', and 'File: Status (Status)', each with an 'Add' button.
- Page settings:** Includes 'Path' (/admin/content/files), 'Menu' (Tab: Files), and 'Access' (Permission: Access the Files overview page).
- Header, Footer, No results behavior:** Each section has an 'Add' button to configure these elements.
- Advanced:** A section containing:
  - Relationships:** Includes 'File usage' and 'User who uploaded' with an 'Add' button.
  - Contextual filters:** Has an 'Add' button.
  - Exposed form:** Includes 'Exposed form in block' (set to No) and 'Exposed form style' (Basic/Settings).
  - Other:** A section for additional configuration options.

# Content Management Systems

## Adding related data in view with relationships



The screenshot shows the Drupal 10 admin interface for the 'Files' view. A modal dialog titled 'Configure Relationship: Broken/missing handler' is open. The dialog contains the following information:

- For:** This page (override)
- Title:** A user that is associated with this file, usually because this file is in a field on the user.
- Relationship:** Do not use a relationship
- Message:** The handler for this item is broken or missing. The following details are available:  
Enabling the appropriate module may solve this issue. Otherwise, check to see if there is a module update available.
- Buttons:** Apply (this display), Cancel, Remove

The background interface shows the 'Files' view configuration page with a sidebar containing 'Displays', 'Title', 'Format', and 'Fields' sections. The 'Fields' section lists fields like 'File: File ID', 'File: Filename', 'File: File MIME type', 'File: File size', and 'File: Status'. The main content area shows 'No results behavior' with an 'Add' button and a 'Global: Unfiltered text' option.

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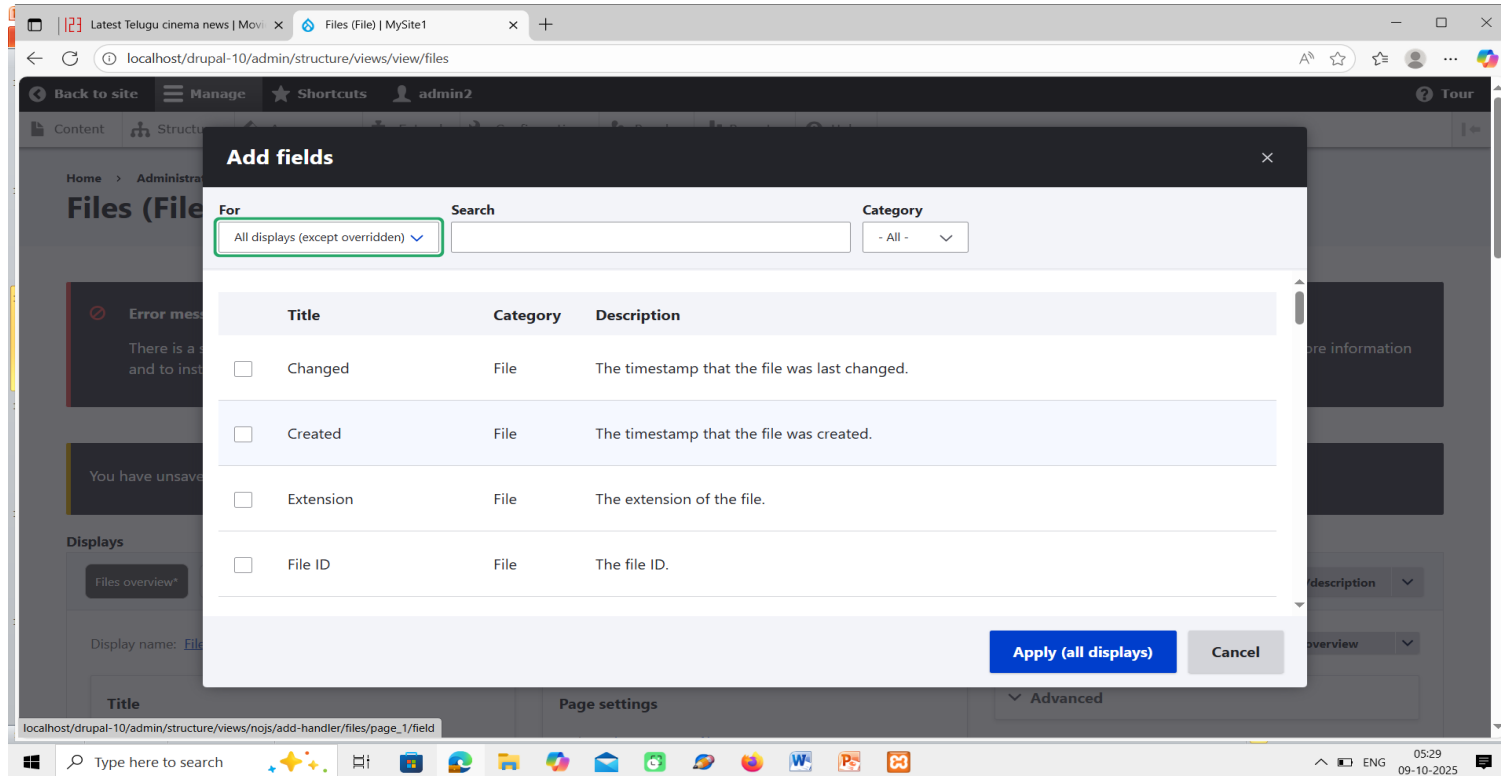
## Adding related data in view with relationships

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6. Add a new field by clicking on Add in the Fields section.
7. Search for extension and select the extension field from the User who uploaded category. Then, click on Apply (all displays):

# Content Management Systems

## Adding related data in view with relationships



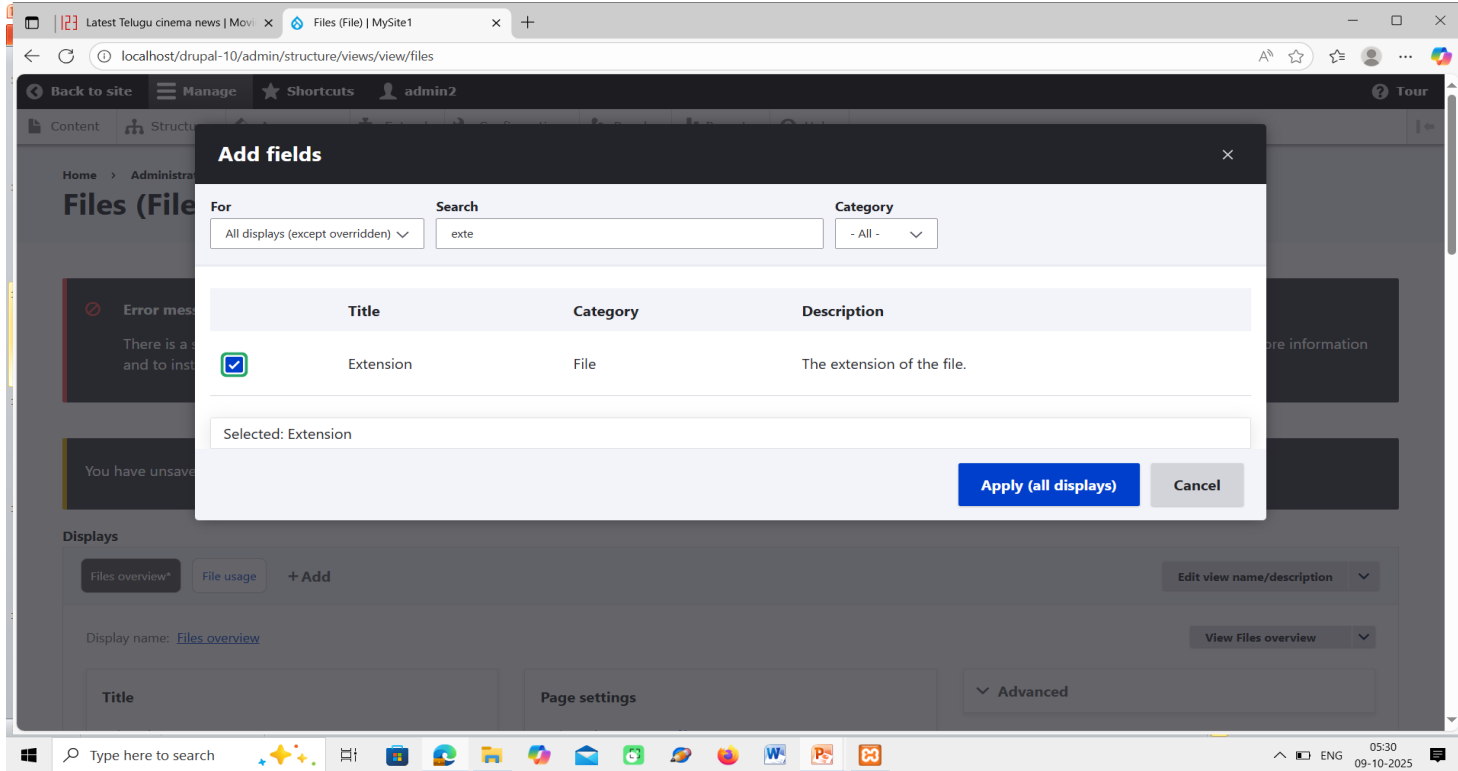
The screenshot shows the Drupal 10 admin interface. A modal dialog titled "Add fields" is open over the "Files (File)" view configuration page. The dialog has a "For" dropdown set to "All displays (except overridden)", a "Search" text field, and a "Category" dropdown set to "All". Below these is a table of available fields:

|                          | Title     | Category | Description                                   |
|--------------------------|-----------|----------|-----------------------------------------------|
| <input type="checkbox"/> | Changed   | File     | The timestamp that the file was last changed. |
| <input type="checkbox"/> | Created   | File     | The timestamp that the file was created.      |
| <input type="checkbox"/> | Extension | File     | The extension of the file.                    |
| <input type="checkbox"/> | File ID   | File     | The file ID.                                  |

At the bottom of the dialog are "Apply (all displays)" and "Cancel" buttons. The background shows the Drupal admin interface with a breadcrumb trail: Home > Administration > Structure > Views > Files (File). The URL in the address bar is localhost/drupal-10/admin/structure/views/view/files. The system tray at the bottom shows the date and time as 05:29 on 09-10-2025.

# Content Management Systems

## Adding related data in view with relationships



The screenshot shows the Drupal 10 administration interface. The browser address bar indicates the URL is `localhost/drupal-10/admin/structure/views/view/files`. The page title is "Files (File) | MySite1". The user is logged in as "admin2".

An "Add fields" dialog is open, showing the configuration for the "Files (File)" view. The dialog has the following fields:

- For:** All displays (except overridden) (dropdown)
- Search:** exte (text input)
- Category:** - All - (dropdown)

The dialog displays a table of available fields:

|                                     | Title     | Category | Description                |
|-------------------------------------|-----------|----------|----------------------------|
| <input checked="" type="checkbox"/> | Extension | File     | The extension of the file. |

Below the table, the "Selected:" field shows "Extension". At the bottom right of the dialog are two buttons: "Apply (all displays)" and "Cancel".

The background interface shows the "Files (File)" view configuration page. It includes a "Displays" section with a "Files overview\*" display and a "File usage" display. The "Display name" is set to "Files overview".

# Content Management Systems

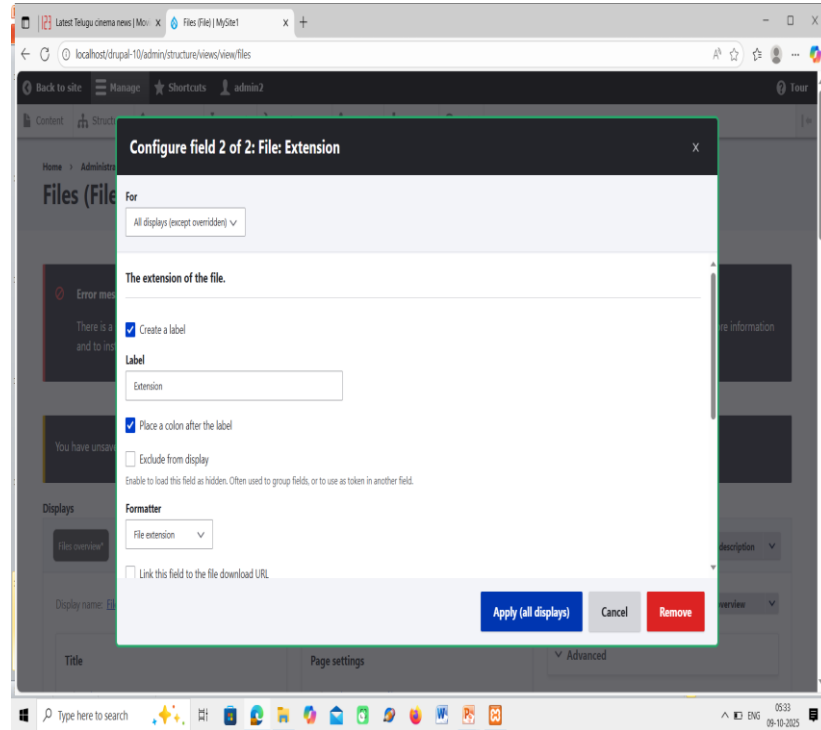
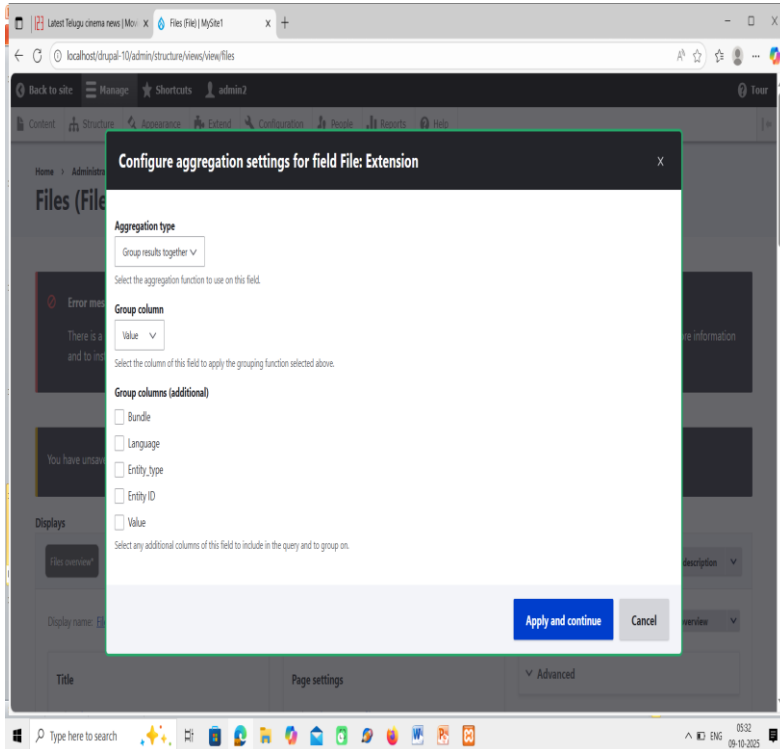
## Adding related data in view with relationships

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8. This view uses aggregation, which presents a new configuration form when you first add a field. Click on Apply and continue to use the aggregation defaults.
9. We will use the default field settings that will provide the label extension. Click on Apply (all displays).
10. Click on Save to finish editing the view and commit your changes.
11. When viewing the files list in `/admin/content/files`, the username of the user that uploaded the file will now be displayed

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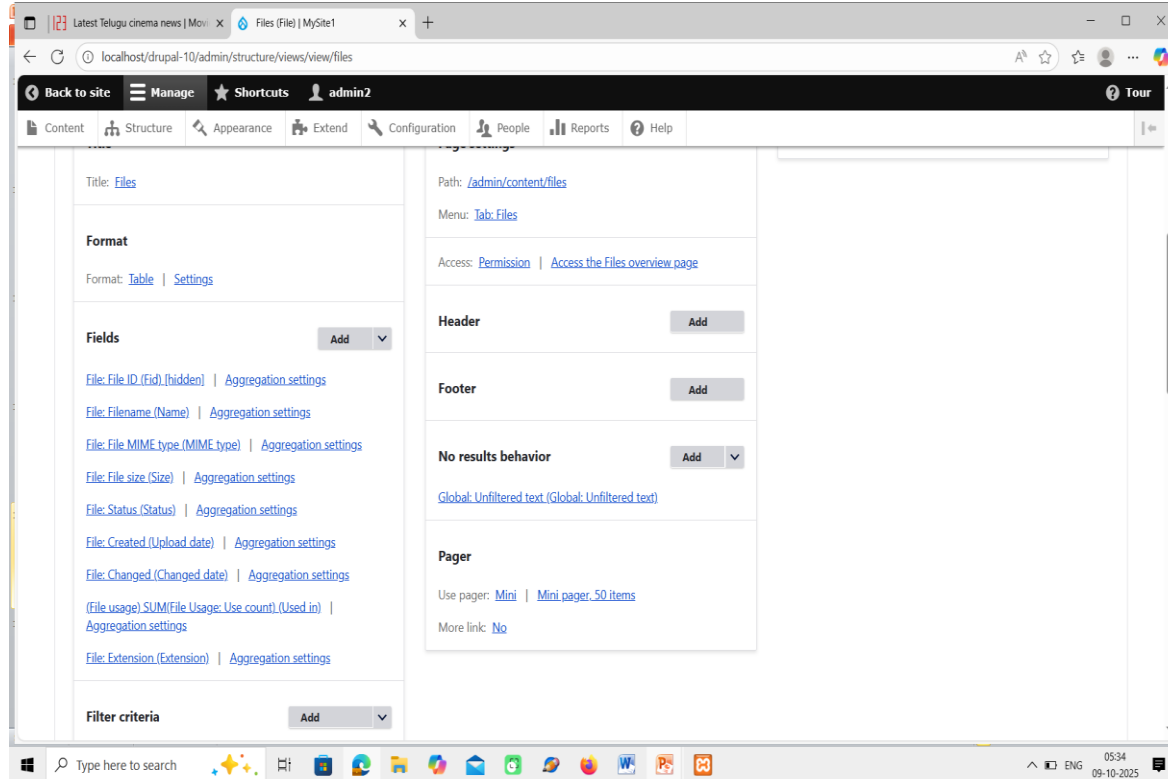
## Adding related data in view with relationships





# Content Management Systems

## Adding related data in view with relationships



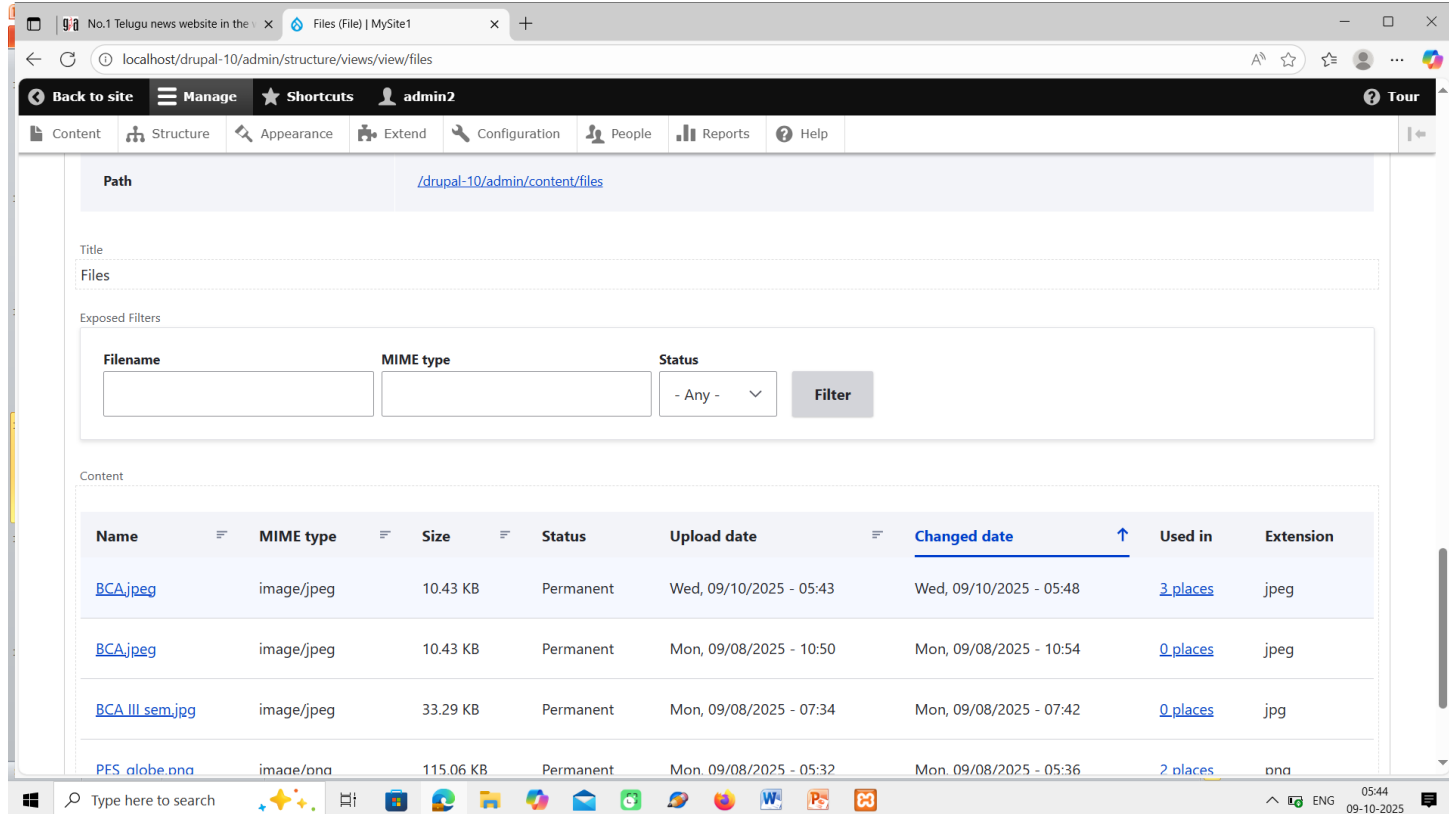
The screenshot displays the Drupal 10 administrative interface for configuring a view named 'Files'. The interface is divided into several sections:

- Title:** Files
- Format:** Table | Settings
- Fields:** A list of available fields for the view, each with an 'Add' button and a link to 'Aggregation settings'. The fields include:
  - File: File ID (Fid) [hidden]
  - File: Filename (Name)
  - File: File MIME type (MIME type)
  - File: File size (Size)
  - File: Status (Status)
  - File: Created (Upload date)
  - File: Changed (Changed date)
  - (File usage).SUM(File Usage: Use count) (Used in)
  - File: Extension (Extension)
- Filter criteria:** A section with an 'Add' button.
- Path:** /admin/content/files
- Menu:** Tab: Files
- Access:** Permission | Access the Files overview page
- Header:** Add
- Footer:** Add
- No results behavior:** Add | Global: Unfiltered text (Global: Unfiltered text)
- Pager:** Use pager: Mini | Mini pager: 50 items. More link: No

The browser's address bar shows the URL: localhost/drupal-10/admin/structure/views/view/files. The Windows taskbar at the bottom indicates the system time as 05:34 on 09-10-2025.

# Content Management Systems

## Adding related data in view with relationships



The screenshot shows the Drupal 10 admin interface for the 'Files' view. The breadcrumb path is [/drupal-10/admin/content/files](#). The view has a title 'Files' and is currently displaying a table of file content. The table includes columns for Name, MIME type, Size, Status, Upload date, Changed date, Used in, and Extension. There are four rows of data, each representing a file named 'BCA.jpeg' or 'BCA III sem.jpg'.

**Path** [/drupal-10/admin/content/files](#)

**Title**  
Files

**Exposed Filters**

| Filename             | MIME type            | Status  |
|----------------------|----------------------|---------|
| <input type="text"/> | <input type="text"/> | - Any - |

**Filter**

**Content**

| Name                            | MIME type  | Size      | Status    | Upload date             | Changed date            | Used in                  | Extension |
|---------------------------------|------------|-----------|-----------|-------------------------|-------------------------|--------------------------|-----------|
| <a href="#">BCA.jpeg</a>        | image/jpeg | 10.43 KB  | Permanent | Wed, 09/10/2025 - 05:43 | Wed, 09/10/2025 - 05:48 | <a href="#">3 places</a> | jpeg      |
| <a href="#">BCA.jpeg</a>        | image/jpeg | 10.43 KB  | Permanent | Mon, 09/08/2025 - 10:50 | Mon, 09/08/2025 - 10:54 | <a href="#">0 places</a> | jpeg      |
| <a href="#">BCA III sem.jpg</a> | image/jpeg | 33.29 KB  | Permanent | Mon, 09/08/2025 - 07:34 | Mon, 09/08/2025 - 07:42 | <a href="#">0 places</a> | jpg       |
| <a href="#">PES globe.png</a>   | image/png  | 115.06 KB | Permanent | Mon, 09/08/2025 - 05:32 | Mon, 09/08/2025 - 05:36 | <a href="#">2 places</a> | png       |

# Content Management Systems

## Adding related data in view with relationships

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How it works...

Drupal stores data in a normalized format. In short, database normalization involves the organization of data in specifically related tables. Each entity type has its own database table, and all fields have their own database table.

When you create a view and specify what kind of data will be shown, you are specifying a base table in the database that Views will query.

# Content Management Systems

## Adding related data in view with relationships

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Views will automatically associate fields that belong to the entity and its relationship to those tables for you.

When an entity has an entity reference field, you can add a relationship to the referenced entity type's table. This is an explicit definition, whereas fields are implicit.

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# Content Management Systems

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## Displaying charts using views

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# Content Management Systems

## Displaying charts using views

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There are many ways to present your data to your readers. One example would be a table or list.

Some times the best approach is to show data on Charts.

It can ease understanding of large quantities of data. There is a way to make charts in Drupal with the help of the Charts module and Views.

# Content Management Systems

## Displaying charts using views

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Charts module is required for creating charts. Charts module offers an API through which charts can be created in a programmatic way.

# Content Management Systems

## Displaying charts using views

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### **Transform data into information**

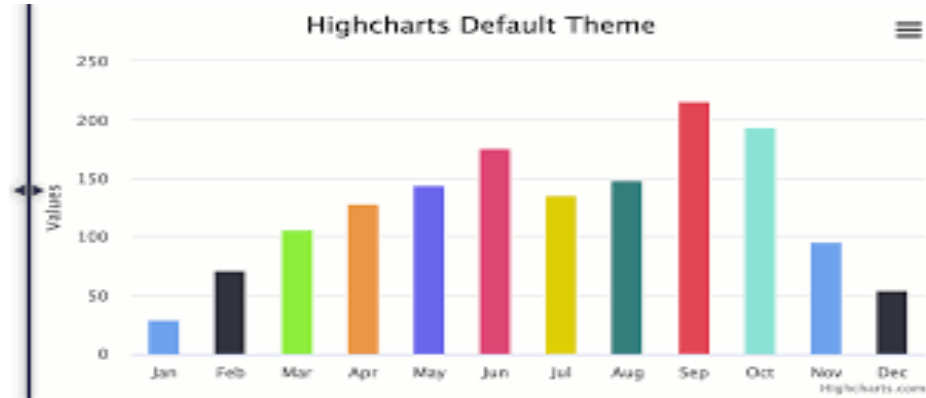
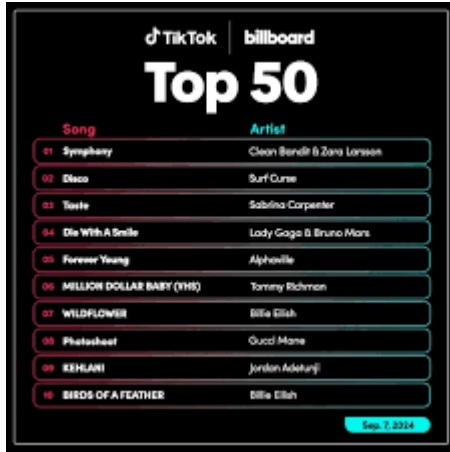
The Charts module enables users to build dynamic charts without writing a line of code by using the Views module or a Chart field (advanced users can make use of the Charts API).

Users can choose between multiple charting libraries, such as Billboard.js, C3.js, Chart.js, Google Charts, or Highcharts. Another important chart that can be implemented is Apex Chart.

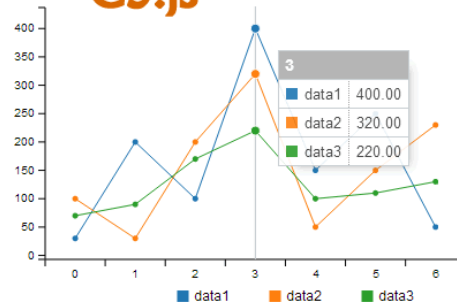


# Content Management Systems

## Displaying charts using views



C3.js



# Content Management Systems

## Displaying charts using views

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Apex charts is a powerful charts library that can be used in Drupal.

To use the Apex charts module in Drupal 10 follow these steps:

- a. Install the Apex charts module from [www.drupal.org](http://www.drupal.org)
- b. (or) give the command Composer require 'drupal/charts-apexcharts'
- c. Enable the module, navigate to Administration>Extend in your Drupal website and enable the charts Apex charts module.

# Content Management Systems

## Displaying charts using views

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Set apex charts as default charting library.

d. Go to Administration>configuration>Content authoring>Charts and select apex charts as your default charting library

e. Create your charts

There are several ways to create charts, using the charts module and apex charts.

### Using views

- create a new view (or) edit existing one.
- change the display format to chart.
- configure the chart settings within the view, including the fields to use for data, axes labels, and chart type.

# Content Management Systems

## Displaying charts using views

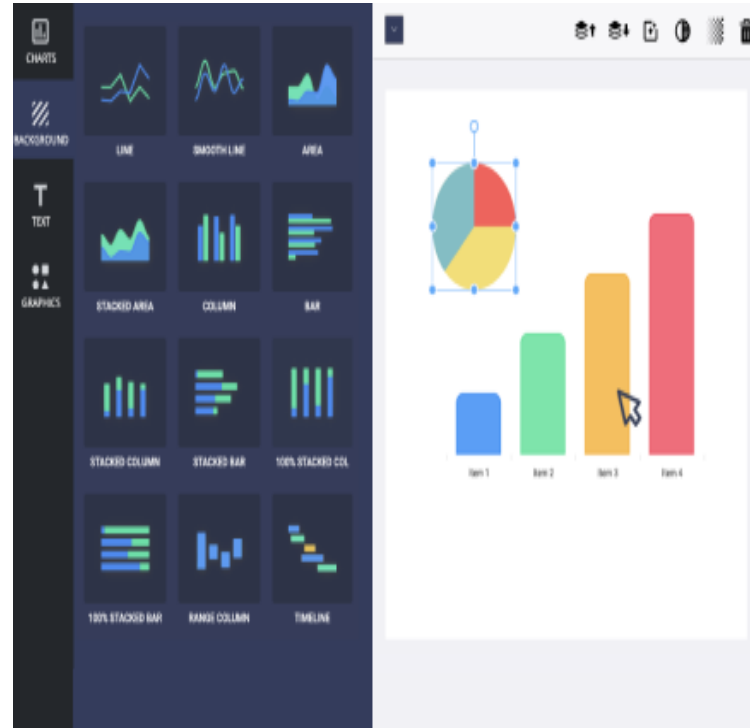
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### Using Charts block

- create a new custom block using Layout builder.
- Add a charts block and configure its settings including the data source, chart type and apex charts specifications.

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## Apex Charts – Dashboard view



# Content Management Systems

## Displaying charts using views

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### Features

**Wide chart type support:** Pie, Line, Bar, Column, Area or Scatter charts.

**API:** A Drupal-friendly syntax for creating charts easily, regardless of the underlying library.

**Views integration:** Display the Views results as charts.

**Examples:** The module includes lots of example charts for you to reference when building your own charts or chart library integration.

**Supported charting libraries:** Out of the box, you may use either [Google Charts](#) or [Highcharts](#) as the charting library.

# Content Management Systems

## Displaying charts using views

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In this step, we will use the Charts module to create graphs using the Views module. The Charts module integrates different graphing libraries with the Views module. The view created in this step will display a graph based on statistics generated from the Statistic modules to graph visits to content on the Drupal site.

Leverage the Charts module to create a view that renders a chart of data.

# Content Management Systems

## Displaying charts using views

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### Getting ready

This step requires having enough data available to put into a graph. We will be using data available from the Statistics module, which tracks content page views.

To generate content and view statistics, we will use the Devel (for Development) module (it may be deprecated after 2022 onwards/it may be least used) This module provides the means of generating sample content and populating statistics.

To use Devel to generate content, we must add it.



# Content Management Systems

## Displaying charts using views

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Now, we can generate content.

Log in to your Drupal site and visit `/admin/config/development/generate/content`.

Check the checkbox for Article to generate articles.

Select no.of nodes as 10, select 1 week ago for how far the nodes should be dated back in time, Give maximum number of comments per node as 1

Rest of all set to default values

Ensure the Add statistics for each node (node\_counter table) checkbox is checked so that statistics are generated. (this option may not be available in Drupal 10)

Press Generate to generate the sample content.

# Content Management Systems

## Displaying charts using views

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How to do it...

1. First, we must add the Charts module to the Drupal site.
2. Go to Structure and then Views. This will bring you to the administrative overview of all the views that have been created. Click on Add view to create a new view.
3. Set View name to Content Statistics and keep the default View settings values.
4. Check the Create a page checkbox to create a page. Use the default values provided. Under Page display settings, change the display format to Chart of fields:

# Content Management Systems

## Displaying charts using views

**Page settings**

☒ Create a page

Page title

Content Statistics

Path

content-statistics

**Page display settings**

Display format: Chart of: teasers

Figure 3.17 – Page settings for the chart view

# Content Management Systems

## Displaying charts using views

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5. Click on Save and edit to continue working on your view.
6. In the Fields section, click on Add. Search for Total views under the Content statistics category. Click on Add and configure fields to continue.
7. Check the Create a label checkbox and set the Label to Total views. Click on Apply to finish adding the field.
8. Next, we need to configure the chart to use our Total views field as the data for the chart. In the Format section, click on Settings next to Chart.
9. Select Google from the Charting library drop-down menu.
10. Keep the Label field option set to Content: Title. In the Provides data column, check the total count checkbox

# Content Management Systems

## Displaying charts using views

**Default: Style options** ✕

**Charting library** \*  

Google ▼

**Grouping field**  

- None - ▼

If grouping by a particular field, that field will be used to determine stacking of the chart. Generally this will be the same field as what you select for the "Label field" below. If you do not have more than one "Provides data" field below, there will be nothing to stack. If you want to have another series displayed, use a "Chart attachment" display, and set it to attach to this display.

☐ Allow advanced rendering  
Allow views field rewriting etc. for label and data fields. This can break charts if you rewrite the field to a value the charting library cannot handle - e.g. passing a string value into a numeric data column.

**Charts fields**  
**Label field**  

☒ Content: Title  
☐ Total views  
☐ No label field

◀ Show row weights

| Field Name     | Provides Data                                  | Color        |
|----------------|------------------------------------------------|--------------|
| Content: Title | <input type="checkbox"/> title                 | <div>I</div> |
| Total views    | <input checked="" type="checkbox"/> totalcount | <div>I</div> |

Apply

Cancel

Figure 3.18 – Chart style settings

# Content Management Systems

## Displaying charts using views

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11. Click on Apply to set the chart settings.
12. Click on Save to save your view.
13. Visit </content-statistics> to see the chart of usage statistics.

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# Content Management Systems

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## Creating a Module

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# Content Management Systems

## Creating a module

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The first step to extending Drupal is to create a custom module. Although the task sounds daunting, it can be accomplished in a few simple steps.

Modules can provide functionalities and customizations for functionalities provided by other modules, or they can be used as a way to contain the configuration and a site's state.



# Content Management Systems

## Creating a module

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In this step, we will create module named `modulename.info.yml` a file containing information that Drupal uses to discover extensions and install the module.

# Content Management Systems

## Creating a module

---

How to do it...

1. In your web/modules directory, create a new directory called custom and then another directory named mymodule inside it. This will be your module's directory.
2. Using the command line, you may create the directory with the following command:

```
mkdir -p web/modules/custom/mymodule
```

This will create the required directories.

3. Create a `mymodule.info.yml` file in your module's directory. This will contain the metadata that identifies the module to Drupal.

4. Add a line to the `mymodule.info.yml` file to provide a name for the module with the name key:

```
name: My Module
```

5. We must define the type of extension with the key type. Drupal does not assume the extension type just by directory location alone:

```
type: module
```

6. The description key allows you to provide extra information about your module, which will be displayed on the module's list page:

description: This is an example module from the Drupal Development Cookbook!

7. Extensions are required to provide a `core_version_requirement` that identifies what versions of Drupal core the module is compatible with, using a semantic versioning constraint:

`core_version_requirement: '>=10'`

Save the mymodule.info.yml file, which looks as follows:

```
name: My Module
type: module
description: This is an example module from the Drupal
Development Cookbook!
core_version_requirement: ^10
Package:Custom
dependencies:
  -drupal:link
  -drupal:views
  -paragraphs:paragraphs
  -webform:webform(>=6.1.0)
```

Give a space after typing name of the module, type, description and core\_version\_requirement

# Content Management Systems

## Creating a module

Next, create a file named mymodule.module in your module file. This is the extension file that will allow us to add hook definitions. It is a regular PHP file, but the file extension matches its extension type of module.

```
<?php

/**
 * Implements hook_page_top().
 */
function mymodule_page_top() {
  \Drupal::messenger()->addStatus('Hello world!');
}
```

# Content Management Systems

## Creating a module

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This implements `hook_page_top`, which is called whenever a page is rendered. It uses the messenger service to add a status message to the page.

Enable the module by installing it through Extend → Add module

Visit your Drupal site. The Hello world! message will be added to each page:

# Content Management Systems

## Creating a module



Figure 4.1 – Drupal page displaying Hello world!



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# Content Management Systems

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## Defining and checking user permissions

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# Content Management Systems

## Defining and checking user permissions

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In Drupal, roles and permissions are used to define robust access control lists for users. Modules use permissions to check whether the current user has access to perform an action, view specific items, or do other operations.

Modules then define the permissions that are used so that Drupal is aware of them. Developers can then construct roles, which are made up of enabled permissions.

In this step, we will define new permission(s) in a module that is used to check if the user can mark content as promoted to the front page or sticky at the top of lists. This permission will be used in an entity field access hook to deny access to the fields if the user is missing the permission.

# Content Management Systems

## Defining and checking user permissions

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### Getting ready

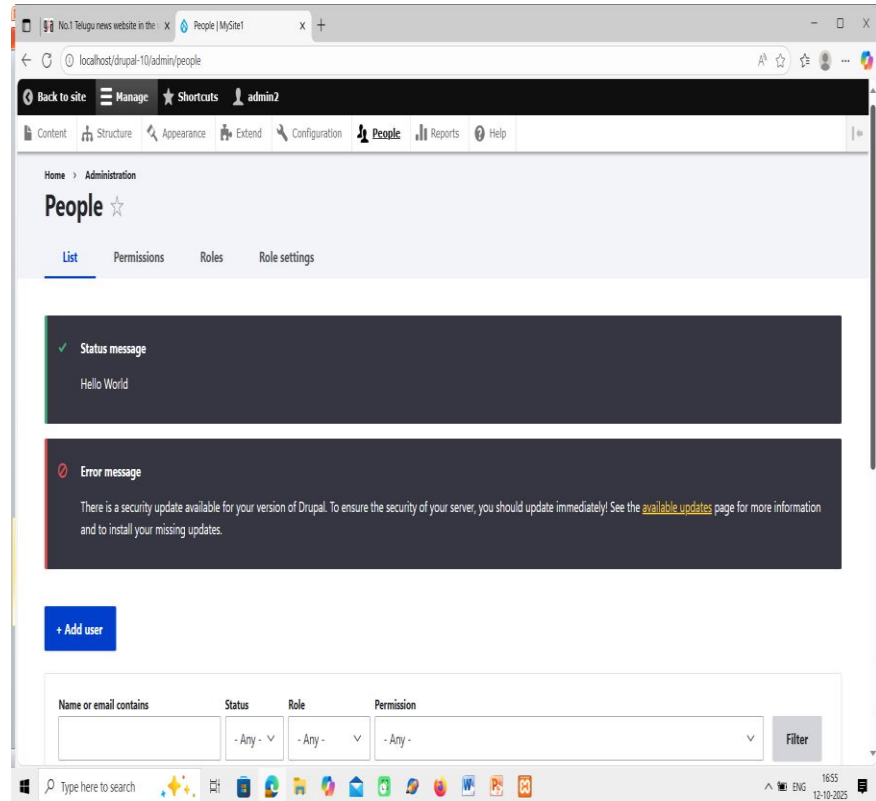
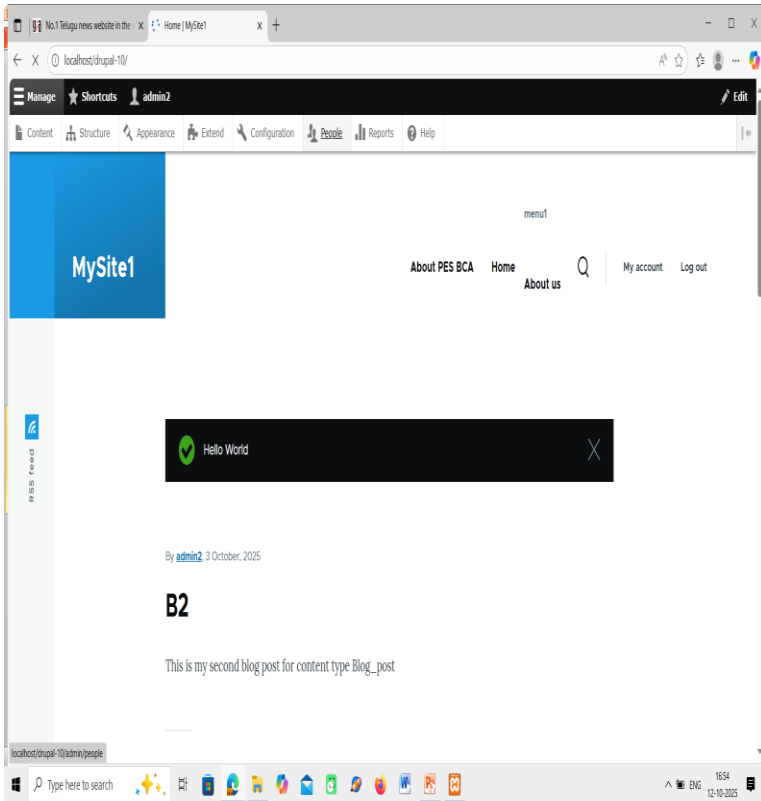
We have already created the module mymodule, create a new user with role content editor.

Go to People→Add User (Enter the email address, username, password and role as content editor, Make the account to be active and timezone to be UTC) and then click on create new account.

Drupal bypasses access checks for the first user. The secondary user will be required to demonstrate the permission.

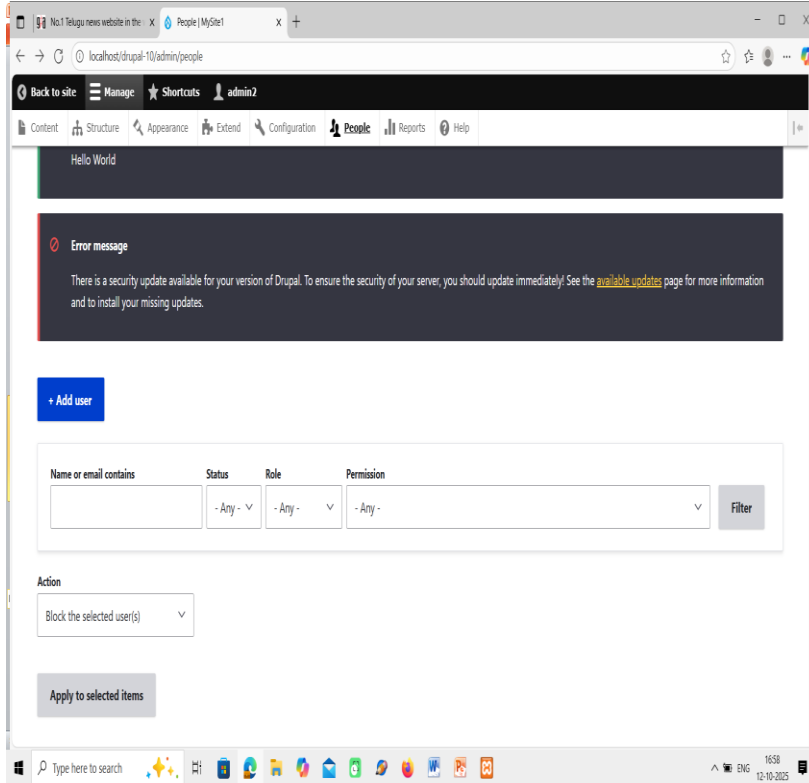
# Content Management Systems

## Defining and checking user permissions



# Content Management Systems

## Defining and checking user permissions



localhost/drupal-10/admin/people

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Content Structure Appearance Extend Configuration People Reports Help

Hello World

**Error message**

There is a security update available for your version of Drupal. To ensure the security of your server, you should update immediately. See the [available updates](#) page for more information and to install your missing updates.

+ Add user

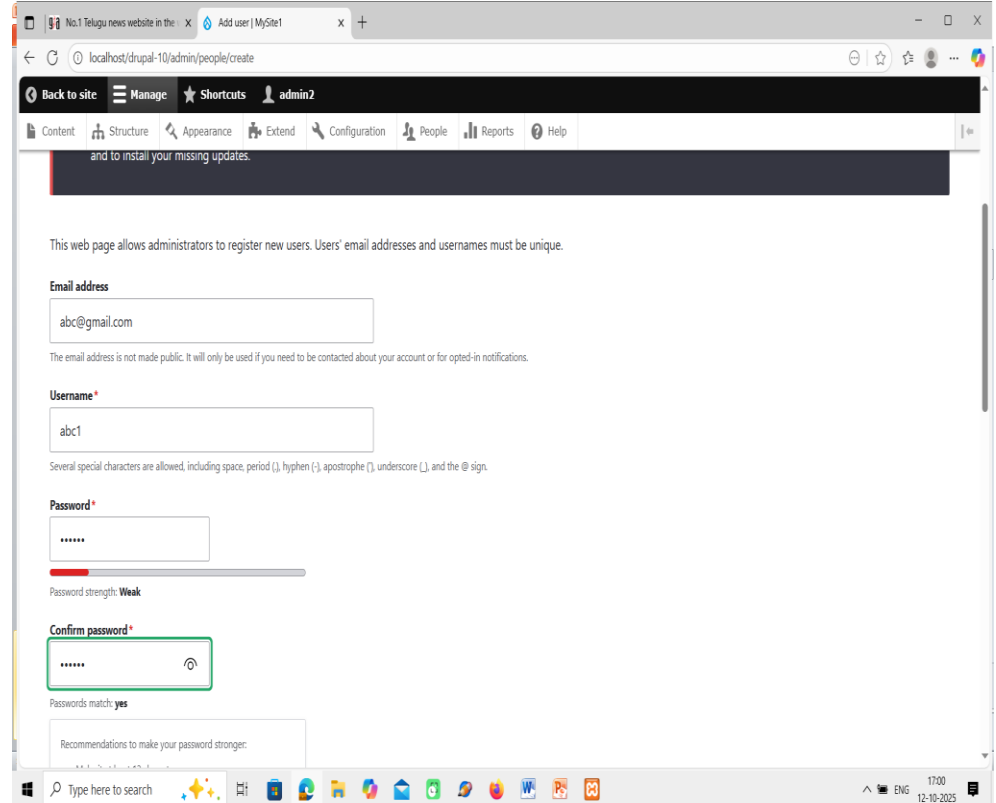
| Name or email contains | Status  | Role    | Permission |
|------------------------|---------|---------|------------|
| <input type="text"/>   | - Any - | - Any - | - Any -    |

Filter

Action

Block the selected user(s)

Apply to selected items



localhost/drupal-10/admin/people/create

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and to install your missing updates.

This web page allows administrators to register new users. Users' email addresses and usernames must be unique.

Email address

abc@gmail.com

The email address is not made public. It will only be used if you need to be contacted about your account or for opted-in notifications.

Username\*

abc1

Several special characters are allowed, including space, period (.), hyphen (-), apostrophe ('), underscore (\_), and the @ sign.

Password\*

.....

Password strength: Weak

Confirm password\*

.....

Passwords match: yes

Recommendations to make your password stronger:

### How to do it...

1. Permissions are stored in a permissions.yml file. Add a mymodule.permissions.yml file to the base directory of your module.
2. First, we will need to define the internal string used to identify this permission, such as can promote nodes:

can promote nodes:

3. Each permission is a YAML array of data. We will need to provide a title key that will be displayed on the permissions page:

can promote nodes:

title: 'Can promote content'

4. Permissions have a description key to provide details of the permission on the permissions page:

can promote nodes:

title: 'Can promote content'

description: 'Determines if the user can change promotion fields on content.'

5. Save your file with `mymodule.permissions.yml` file, and edit the module's `mymodule.module` file so that you may write the hook to use this permission.

6. In your mymodule.module file, add a function named mymodule\_entity\_field\_access to implement hook\_entity\_field\_access. This is invoked to control access at a granular level per field on entity forms:

```
function mymodule_entity_field_access(  
  $operation,  
  \Drupal\Core\Field\FieldDefinitionInterface  
  $field_definition,  
  \Drupal\Core\Session\AccountInterface $account  
) {  
  return \Drupal\Core\Access\AccessResult::neutral();  
}
```

Drupal uses access result value objects to handle access results. Access results may be neutral, forbidden, or allowed. Hooks implementing hook\_entity\_field\_access must return an access result and cannot return null.



# Content Management Systems

## Defining and checking user permissions

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7. In our hook, we will check the field name being checked. If the field name is promote or sticky, we will check whether the user has the can promote nodes permission and return that access result:

```
function mymodule_entity_field_access(  
  $operation,  
  \Drupal\Core\Field\FieldDefinitionInterface  
  $field_definition,  
  \Drupal\Core\Session\AccountInterface $account  
) {  
  $field_name = $field_definition->getName();  
  if ($field_name === 'promote' || $field_name ===  
  'sticky') {  
    $can_promote_nodes = $account->hasPermission('can promote nodes');  
    return \Drupal\Core\Access\AccessResult::allowedIf  
    ($can_promote_nodes);  
  }  
  return \Drupal\Core\Access\AccessResult::neutral(); }  

```

# Content Management Systems

## Defining and checking user permissions

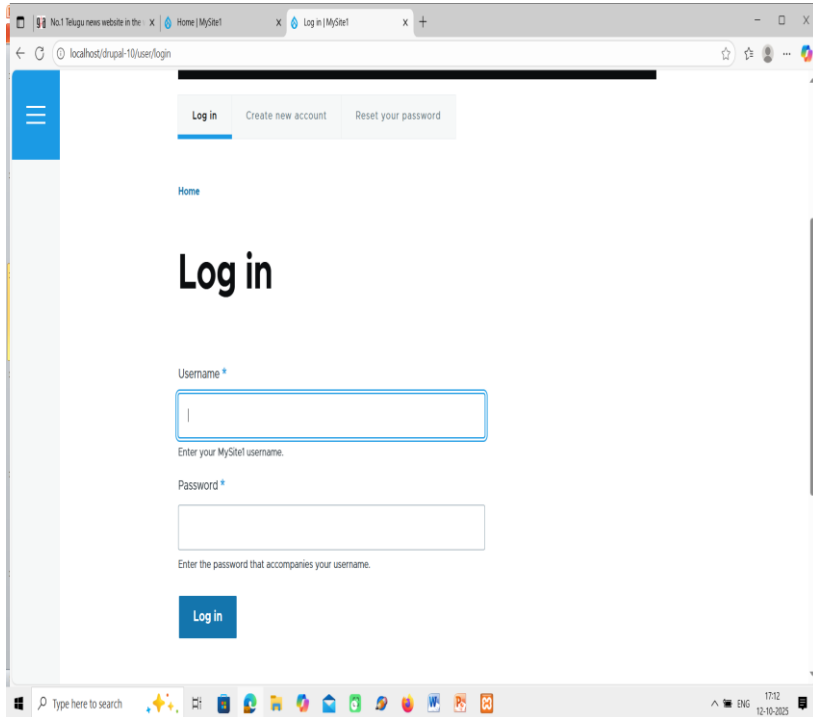
---

The access result object has an `allowedIf` method that returns an appropriate result based on the parameter provided. In this case, it will return `AccessResult::allowed()` if the user has the permission or `AccessResult::neutral()` if not.

8. New permissions are not granted to roles automatically. In another browser or guest tab, log in as a user with the Content editor role and create a piece of content. The Promotion options section of the sidebar will be missing since we do not have field access:

# Content Management Systems

## Defining and checking user permissions



The screenshot shows a web browser window with the address bar displaying 'localhost/drupal-10/user/login'. The page has a blue sidebar on the left with a hamburger menu icon. At the top, there are three tabs: 'No.1 Telugu news website in the...', 'Home | MySite1', and 'Log in | MySite1'. Below the tabs, there are three buttons: 'Log in' (active), 'Create new account', and 'Reset your password'. The main content area has a 'Home' link and a large 'Log in' heading. Below the heading, there are two input fields: 'Username \*' and 'Password \*'. The 'Username \*' field has a placeholder text 'Enter your MySite1 username.' and the 'Password \*' field has a placeholder text 'Enter the password that accompanies your username.' Below the password field is a blue 'Log in' button. The Windows taskbar is visible at the bottom with the search bar and various application icons.

# Content Management Systems

## Defining and checking user permissions

Home > Add content

### Create Article

**Title \***

**Body ([Edit summary](#))**

**B** ***I***       **Format**   **Source**

[About text formats](#)

**Last saved:** Not saved yet

**Author:** editor

**Revision log message**

Briefly describe the changes you have made.

✓ **URL alias**  
No alias

Figure 4.2 – Promotion options are hidden due to missing permissions

# Content Management Systems

## Defining and checking user permissions

---

9. As your administrative user, go to People and then to Permissions to add your permission for the Content editor role in the My Module section:

# Content Management Systems

## Defining and checking user permissions

| Permission                                                                            | Content editor                      |
|---------------------------------------------------------------------------------------|-------------------------------------|
| <b>My Module</b>                                                                      |                                     |
| Can promote content<br>Determines if the user can change promotion fields on content. | <input checked="" type="checkbox"/> |

Figure 4.3 – Adding the permission to the Content editor role

# Content Management Systems

## Defining and checking user permissions

10. Using your Content editor user, try and create a piece of content once more. The Promotion options section in the sidebar will be present:

Home > Add content

### Create Article

**Title \***

**Body (Edit summary)**

**B** *I*          Format -  Source

[About text formats](#)

**Tags**

**Last saved:** Not saved yet  
**Author:** editor

**Revision log message**

Briefly describe the changes you have made.

**URL alias**  
No alias

**Promotion options**  
Promoted to front page

Figure 4.4 – The Promotion options section appears once the permission has been granted

# Content Management Systems

## Defining and checking user permissions

---

How it works...

Permissions and roles are provided by the User module. The user.permissions service discovers the permissions.yml file provided by installed modules.

By default, the service is defined through the \Drupal\user\PermissionHandler class. Drupal does not save a list of all available permissions. The permissions for a system are loaded when the permissions page is loaded. Roles contain an array of permissions.



# Content Management Systems

## Defining and checking user permissions

---

When checking a user's access for permission, Drupal checks all the user's roles to see whether they support that permission.

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# Content Management Systems

---

## Creating event subscriber to react to events

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Department of Computer Applications

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# Content Management Systems

## Creating event subscriber to react to events

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Drupal has two ways of integrating various parts of the system: using hooks or events. Hooks have been a part of Drupal for its entire lifespan and events were introduced in Drupal 8.

Unlike the hook system, which has implicit registration, the event dispatch system uses explicit registration for an event.

# Content Management Systems

## Creating event subscriber to react to events

---

The events dispatcher system comes from the Symfony framework and allows components to easily interact with one another.

Within Drupal, and integrated Symfony components, events are dispatched, and event subscribers can listen to the events and react to changes or other processes.

In this step, we will subscribe to the RequestEvent event, which fires when a request is first handled. If the user is not logged in, we will navigate them to the login page.

# Content Management Systems

## Creating event subscriber to react to events

---

How to do it...

1. Create src/EventSubscriber/RequestSubscriber.php in your module.
2. Define the RequestSubscriber class, which implements the EventSubscriberInterface interface:

```
<?php

namespace Drupal\mymodule\EventSubscriber;

use Symfony\Component\EventDispatcher\EventSubscriber
    Interface;

class RequestSubscriber implements EventSubscriber
    Interface {

}
```

3. To satisfy the interface requirements, we must add a `getSubscribedEvents` method. This tells the system which events we are subscribing to and the method that needs to be invoked:

# Content Management Systems

## Creating event subscriber to react to events

```
<?php

namespace Drupal\mymodule\EventSubscriber;

use Symfony\Component\EventDispatcher\EventSubscriber
    Interface;
use Symfony\Component\HttpKernel\Event\RequestEvent;

class RequestSubscriber implements EventSubscriber
    Interface {

    /**
     * {@inheritdoc}
     */
    public static function getSubscribedEvents() {
        return [
            RequestEvent::class => ['doAnonymousRedirect',
                28],
        ];
    }
}
```

# Content Management Systems

## Creating event subscriber to react to events

---

Event names are derived from the PHP class name for the event object. In the `getSubscribedEvent` method, we construct an associative array to be returned.

The event class name is the key and our class method to invoke when that event is dispatched.



# Content Management Systems

## Creating event subscriber to react to events

4. Create the doAnonymousRedirect method we specified, which will receive the RequestEvent object for the current request:

```
<?php

namespace Drupal\mymodule\EventSubscriber;

use Drupal\Core\Url;
use Symfony\Component\EventDispatcher\EventSubscriber
    Interface;
use Symfony\Component\HttpFoundation\RedirectResponse;
use Symfony\Component\HttpKernel\Event\RequestEvent;

class RequestSubscriber implements EventSubscriber
    Interface {

    /**
     * Redirects all anonymous users to the login page.
     *
     * @param \Symfony\Component\HttpKernel\Event\
     *         RequestEvent $event
     *     The event.
     */
```

# Content Management Systems

## Creating event subscriber to react to events

```
public function doAnonymousRedirect(RequestEvent
    $event) {
    // Make sure we are not on the user login route.
    if (\Drupal::routeMatch()->getRouteName() ==
        'user.login') {
        return;
    }
    // Check if the current user is logged in.
    if (\Drupal::currentUser()->isAnonymous()) {
        // If they are not logged in, create a redirect
        response.
        $url = Url::fromRoute('user.login')->toString();
        $redirect = new RedirectResponse($url);
        // Set the redirect response on the event,
```

# Content Management Systems

## Creating event subscriber to react to events

```
        canceling default response.
        $event->setResponse($redirect);
    }
}

/**
 * {@inheritdoc}
 */
public static function getSubscribedEvents() {
    return [
        RequestEvent::class => ['doAnonymousRedirect',
            28],
    ];
}
}
```

To prevent a redirect loop, we will use the RouteMatch service to get the current route object and verify that we are not already on the user.login route page.

Then, we check whether the user is anonymous and, if they are, set the event's response to a redirect response.

5. Now that we have created our class, create a mymodule.services.yml file in your module's directory.

6. We must register our class with the service container so that Drupal recognizes that it will act as an event subscriber:

# Content Management Systems

## Creating event subscriber to react to events

---

```
services:
  mymodule.request_subscriber:
    class: Drupal\mymodule\EventSubscriber\
      RequestSubscriber
    tags:
      - { name: event_subscriber }
```

The event\_subscriber tag tells the container to invoke the getSubscribedEvents method and register its methods.

# Content Management Systems

## Creating event subscriber to react to events

---

7. Install the module or rebuild Drupal's caches if it has been already installed.
8. Navigate to any page as an anonymous user – you will be redirected to the login form.

# Content Management Systems

## Creating event subscriber to react to events

---

How it works???

Throughout Drupal and Symfony, component events can be passed to the event dispatcher.

The `event_dispatcher` service in Drupal is an optimized version of the one provided by Symfony but is completely interoperable and provides backward compatibility layers with Symfony.

# Content Management Systems

## Creating event subscriber to react to events

---

When the container is built, all services tagged as `event_subscriber` are gathered.

They are then registered into the `event_dispatcher` service, keyed by the events returned in the `getSubscribedEvents` method.



# Content Management Systems

## Creating event subscriber to react to events

---

When the `event_dispatcher` service is told to dispatch an event, it invokes the registered methods on all subscribed services.

Drupal still primarily uses named events over event objects, as many events leverage the same event object class.

# Content Management Systems

## Creating event subscriber to react to events

---

The `\Symfony\Component\HttpKernel\KernelEvents` class documents the events available to interact with the request life cycle to become a response and even after the response is sent, as we did with `RequestEvent`.



**THANK YOU**

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